Two Step

"Your journey, two steps at a time"

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1. Executive Summary

1.1 The Problem

As society becomes increasingly centred on the online and digital space, we are losing our natural curiosity for the world around us. Physical stores are closing as e-commerce takes over. Our daily lives are being built around efficiency rather than wellbeing. There is so much information immediately available to people via their mobile phones that the sense of joy when we learn new information has been cheapened [1] .

We are also bombarded with so many new inputs everyday. People are overwhelmed and no longer have the capacity or desire to search for new places. COVID resulted in a rise of burnout and disconnection from our surroundings [2]. It is more important than ever to slow down and look for enjoyment in the everyday.

1.2 The Solution

Two Step offers an alternative navigation method to recapture that lost joy of exploration. We have built a platform that bridges the gap between our online world and the areas we live in, bringing back the excitement of discovering hidden gems. With Two Step, users can plan routes not purely on speed or cost but instead prioritising taking a scenic route. Cafes, parks, pubs and more are featured so no matter the time or occasion there is somewhere and something to do near you.
This system is available for both IOS and Android devices. It will initially be rolled out to London then expanded much like Citymapper. We believe Two Step will spark a new wave of experience focused travel and become the leader in the new slow living movement.

Figure 1 - Two Step app user experience: from route generation to gamification

1.3 The Design Process

Stakeholders were contacted throughout the development stage to ensure appropriate and informed decisions were made. End users were contacted initially via a survey to build a design specification which was further validated through initial surveys to curate personas. Finally once a prototype was developed, 12 potential users were approached whilst out walking to validate the overall app experience.

Our other key stakeholder is local business owners who want to advertise through the app. Conducting online research proved there is appetite for this opportunity. Of small business surveyed, 94% planned to increase their marketing budget during 2024, building on the 5-10% of revenue currently being put towards marketing [3]. Once a solid UI/UX was built, business owners and managers were contacted to validate they would pay to be featured and promoted through Two Step.

1.4 Our Concept

Our app uses open source data alongside the Google Maps API to analyse places of interest and build custom routes around the city based on user preferences. Additional gamification and use of the mascot Twoto build user engagement and create joy in the everyday. The Two Step algorithm selects POI in accordance with user inputs and plots the rout one place at a time using the Navigation API until the desired walking duration has been met. Speaking to an experienced DevOps Engineer gave us confidence in the feasibility of this system.

1.5 Value Proposition

Two Step offers unique value to users as a novel app targeting those living in urban environments who wish to explore new spots without needing to wade through endless recommendation sources. Two Step also provides valuable marketing and promotion opportunities for local businesses.

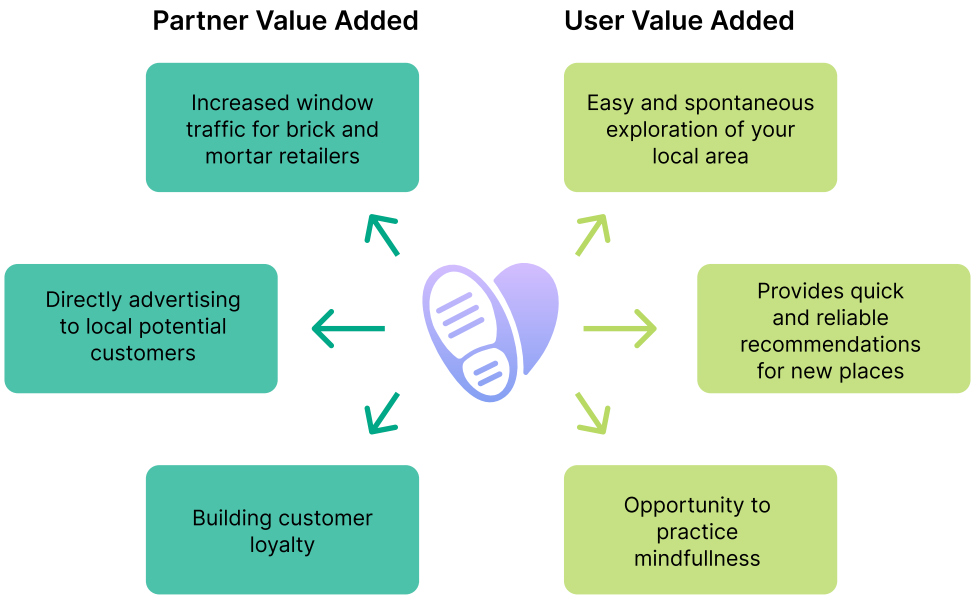


Figure 2 - Summary of the Two Step Value Proposition

1.6 Business Model

Whilst it is challenging to reliably predict the available market size for a relatively novel app concept, we predict a 5 year SOM of **£9.13 million** after analysing top down, bottom up, and value theory market size calculations.

Two Step's business model utilises 4 key revenue streams:
1. Partnerships with local businesses
2. PPC Ads on the app
3. User subscription fees
4. Business exposure

Assumptions:
Across all assumptions made in this analysis, the three that posed the most risk were identified as:
1. Users trusting the app's ability to protect their data and maintain privacy
2. Mapping data is accurate and readily available
3. Generating randomised yet customised routes is technically feasible

Once identified, key steps were put in place to mitigate potential threats to the financial success of Two Step.

Sales Channels:
Our main marketing focus will be on social media engagement through PPC ads with Meta. The majority of UK adults interact with at least one Meta product per day enabling us to reach a large potential user base. A further 25% of marketing will be allocated to app store advertising as we know our target audience is dissatisfied with the current options available and will be searching for alternatives. Another 15% of the marketing strategy will be put towards in person banner ads placed around central London to reach those already walking in the city but looking for more direction. The final component of marketing will be made up of organic word of mouth conversations and individual users sharing their journeys and experiences online.

1.7 Profit/Loss Statement

Two Step is estimated to need £582k of initial investment to cover the initial set up and development costs. Whilst this is higher than ideal the long term projections for profit are promising. After breaking even at 54 months, Two Step is estimated to be worth over £250k 5 years from starting. Given the general positive trends for apps within the navigation space it is likely to assume this value will only rise with time.

2. Industry Opportunity Identification

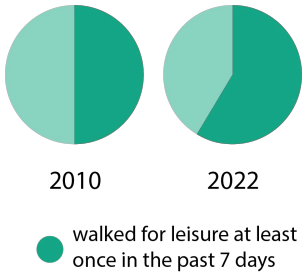


Figure 3 - US walking trends, 2010 [4] and 2022 [5]

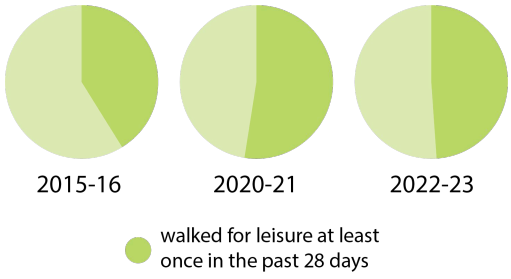


Figure 4 - Uk walking trends [6]

Despite volatility post COVID-19, the navigation app market has been rising with the growing usage and reliance on mobile devices. Additionally COVID-19 lockdowns caused a significant increase in people walking for leisure [6][7]. Whilst not all those who started walking in COVID have continued, there is still an overall rise in popularity in both the UK and US.

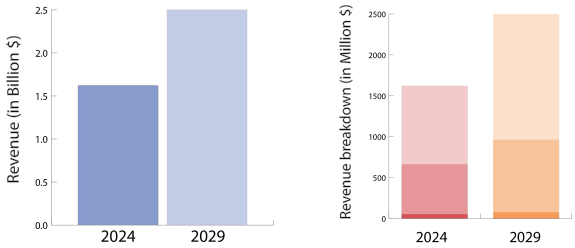


Figure 5 - Market Revenue Predicted Growth [8]

We are also seeing a growing shift in revenue type proportions. Advertising has risen more significantly than either in-app purchasing or initial app purchase [8] and by 2029 is predicted to account for over 60% of total revenue.

2.1 Wide market trends

P	E	S	T	E	L
Walkability grants for urban infrastructure development [9] AI concerns lead to tighter data protection regulations [10] Council grants to catalyse tourism [11]	Cost of living crisis continues impacting households [12] Tourism and travel contributed 9.1% of global GDP in 2023, predicted by the WTTC to rise to 10% in 2024 [13]	**Rise of slow and local travel [14]** 73% of UK adults visited Google Maps in 2024 [15] **Social media groups share top tourist spots [16]** Rise of working from home and flexible hours [17]	New platforms provide ability to navigate in areas with limited or denied GPS signals [18] New apps using AR to display navigation paths over real world view [19]	**Pressure on tourism industry to reduce carbon emissions [20]** Consumers want transparent and sustainable practices from businesses [21] Optimal routes used to reduce pollution [22]	Strict privacy policies for apps on both IOS and Android. Users must be informed what data is collected and why [23] **More EU cities creating clean air/ low emission zones [24]**

Figure 6 - PESTEL analysis of wider trends impacting the navigation app space

Key drivers for our business opportunity were identified as followed:

1. Walking replacing driving in daily travel

When observing how infrastructure development is changing - particularly in built up cities - there are increasing policies setting out the need to develop with pedestrian transport in mind. The expansion of the ULEZ in London and creation of Zero Emission Zones such as in Oxford show the UK's intention to discourage

powered vehicle usage. In addition many employees in the UK are now on flexible contracts and as such no longer need to drive to work. Instead they are more likely to talk shorter walks daily for shopping or leisure and move away from a reliance on cars. Road only navigation apps such as Waze will become obsolete leaving space for alternatives which exclusively aid walking.

2. Looking for joy closer to home

Given the turbulent and challenging past 5 years, many users on social media have formed communities sharing small moments of enjoyment through recommending their favourite spots close to home, both for locals and tourists. In addition following lockdowns, reduced financial freedom and growing understanding of climate change, many are choosing to take smaller scale "slow" holidays within their own country. This means there are more people than ever searching for unique and exciting spots close to their own location or holiday destination. Whilst there are several fragmented online sites for reviews there is space for a purpose build navigation app that not only suggests but directs users to points of interest.

2.2 Navigation app SWOT analysis

Completing a SWOT analysis support our key opportunity of promoting wellbeing and sustainability within the navigation app space. Additionally the potential for partnering with local businesses was highlighted as a form of revenue generation. The diversity of revenue for apps and brand partnerships leads to a strong and flexible funding stream which is capable of adapting to a changing economic or social landscape.

Building on the current strengths there are numerous existing algorithms and optimisation strategies which suggest recommended business advertisements to users on their journeys. This can be taken further to construct routes which direct users specifically to locations or stores the analytic tools suggest will be enjoyable to each individual.

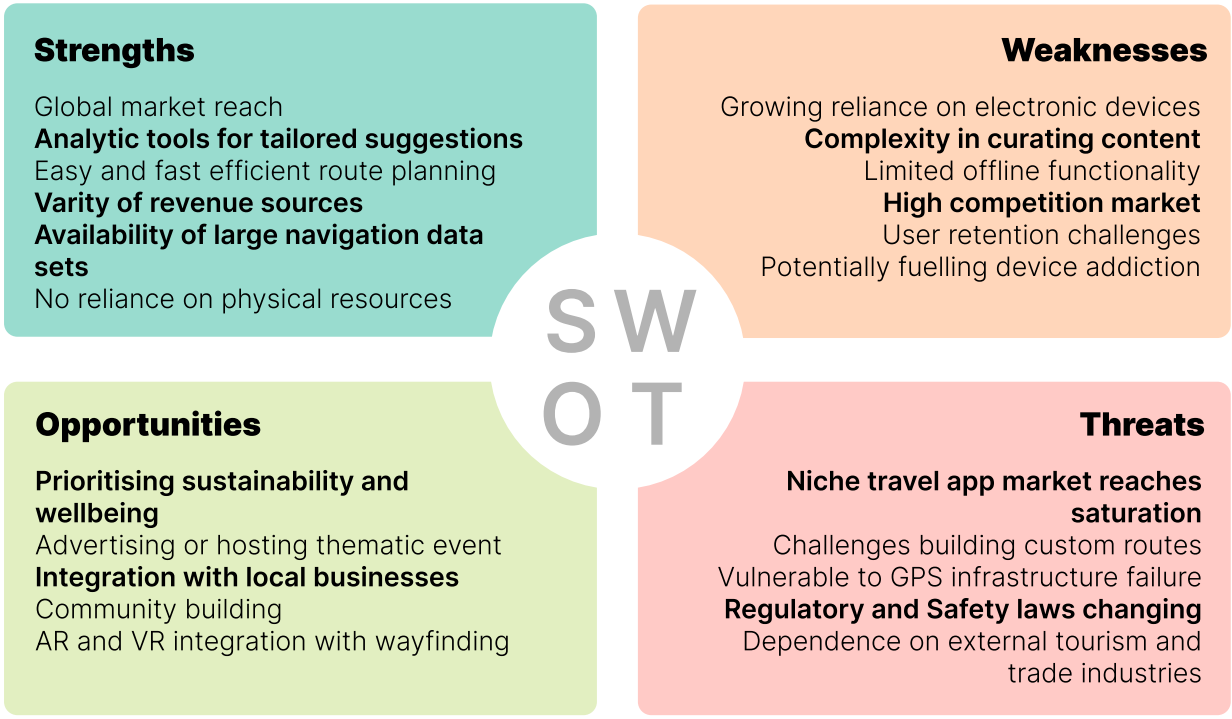


Figure 7 - SWOT analysis of the navigation app market

The key challenges have also been identified for our product. It is likely to assume others will also attempt to fill this gap in the current market meaning competition levels will rise, with potential saturation of the market. Building unique and hard to replicate features will ensure distinction in the case of a crowded field.

The changing regulatory legislation may cause concern for exiting and new navigation apps should GPS data become strictly controlled so any new apps should not rely on tracking or storing route data. To this end, the ideal features would use generic map data to generate routes combined with live GPS tracking for aiding navigation however all user location data will be deleted and not stored in any centralised location. Any saved route paths will be of algorithm generated journeys not representative of actual user movements.

2.3 Competitor landscape

Direct competitors:
Other apps which have the specific focus of navigation are our main competitors. Google maps is by far the most popular and well known navigation app and comes as standard on all Android phones making it the most significant competitor. Citymapper is only rolled out across specific cities so doesn't have a large market share however within those cities it sees high usage percentages. Strava and Komoot were also considered direct competitors as they curate and share route maps for hikers and runners. Finally, Ordinance Survey physical maps and guidebooks provide the analogue competitor to navigation apps.

	User price	Revenue streams	Target persona	User experience	Available platforms
Google Maps	Free	→ PPC ads → Pay to push your business to top of search results → Sell access to APIS for developers	General individual users	→ Search for category of place/ specific destination → Given generated routes → Select and start route → Live visual directions → Optional audio guiding	IOS and Android mobile apps for smartphones and wearable devices HTTPs web platform
Citymapper	Free with ads $1.49/month or $9.99/year without ads	→ User subscriptions → Referral fees when booking taxis in app → Parent company Via use data to make and sell transport tools	People living in or visiting a large city	→ Home page has nearby tube and bus departures → Select destination → Given generated route → Select and start route → Can see gamified stats	
STRAVA	Free with limited tools Full access is £8.99/month or £54.99/ year	→ User subscriptions → Partnerships with external companies to sponsor challenges → Selling data to city planners	Athletes tracking workouts, especially runners or cyclists	→ Route planning via web → View other users routes → Start run and follow direction off phone screen → Share with friends → See and analyse stats	
komoot	Free base app Premium £59.99/year or OTP for specific map areas	→ User subscriptions → One off area maps/ bundles	People who enjoy exercising outdoors, especially hikers	→ Input target destination → Gives route options following cycle paths → Option to change default to city road cycling → Share activity to friends	
Ordnance Survey	Explorer maps £12.99 per region specific area guide book £7.99 - £12.99	→ Paper maps (only 6% of total revenue [25]) → Selling data licences → Government advice → Digital subscription to OS maps	Individuals who prefer offline navigation Agencies	→ Choose area and buy appropriate map → Plan route yourself according to your needs → Rely on landmarks for navigation in transit	Buy online direct from OS or from in person book or travel stores

Figure 8 - Competitor mapping

Key learnings:
The majority of competitors provide a mobile app accompanied by a website strengthening, our assumption that paper based navigation is not a growing market space. In addition, most use a freemium model and rely on subscriptions for a significant portion of revenue. Whilst selling data is a common practice we aim to differentiate ourselves here by committing to user privacy and data security. This further distinguishes Two Step as a wholly wellbeing focused app, placing the desires and security of users over increased potential profits. In terms of targeting users, there is a lot of attention given to “outdoorsy” people and “city” people however little is done to address those who either fall in between or into both groups. A key retention factor for smaller apps having to compete with large international platforms such as Google Maps is novelty and gamification which are both provided through the Two Step user experience.

Indirect competitors:
Whilst not navigation focused, there are other apps and experiences which aim to provide the same exploration and joy in transport. The most culturally significant of these is Pokémon Go, having started a walking phenomenon which boosted overall health, wellbeing and connection worldwide [26].

Key gamification successes:

1. Badges and rewards as you complete more of the game
2. Users join teams and battle against other groups bringing a feeling of community and connection with others
3. AR interactions with characters brings the experience to life

As an app which emphasises exploring your local area on foot there is a large cross over of added value. However Pokémon Go is strongly rooted in its fictional world rather than promoting or directing users to real world end destinations like shops or cafes. In this sense Two Step has something new to offer. In addition, Pokémon Go whilst still having a active and passionate user base has experienced a large decline in overall users and revenue [27]. A large contributing factor was the increase of in app purchase prices for items such as PokéCoins and Remote Passes. Additionally whilst Niantic has regularly updated the app to include further generations of Pokémon, it has lost the attention of many only retaining those particularly dedicated users. This demonstrates the need for continual development to reduce risk of losing users.

A second indirect competitor is GooseChase, a platform for organisers to build scavenger hunts for a group of people, commonly for on boarding, team building or campus orientations. This experience has the same end goal of walking around an area searching for interest however is aimed towards companies or one person building an experience for many others rather than personal exploration.

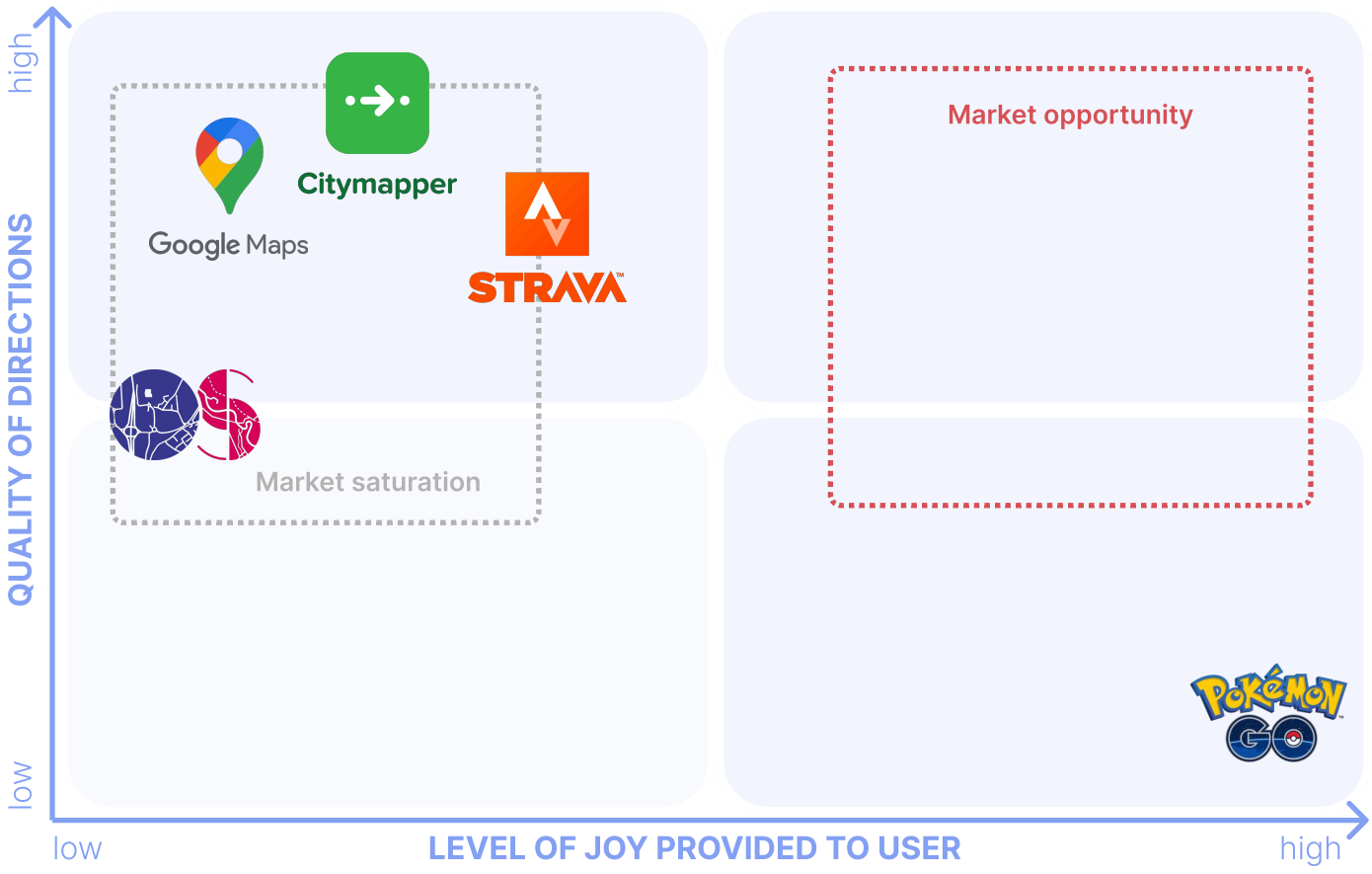


Figure 9- Competitor experience mapping

2.4 Design Opportunity Analysis

Following market analysis there were several gaps identified which inform our key paint points and core users:

1. Apps focusing on city transport only emphasis efficiency
2. Apps which highlight exploration are focused on hiking paths only
3. Hard to find new places of interest in your area

Market need: Low effort exploratory navigation.

Stakeholder	Challenge	Solution
Casual Strollers	Finding destinations when no concrete idea is in mind.	Suggesting a journey based on local events and points of interests.

In addition, physical retailers and other localities struggle to find customers in an e-commerce dominated industry [25]. Many rely on an online presence, where advertising doesn't consider specific location and instead favours large franchises. Unless shops are located in an area of high pedestrian traffic, they can't rely on people passing by to discover them.

Market need: Increased window traffic.

Stakeholder	Challenge	Solution
Physical stores	Getting people to walk past the stores	Direct foot traffic past business locations and points of interest

Market need: Facilitating sharing location data to potential costumers.

Stakeholder	Challenge	Solution
POI advertisers	Maps don't effectively find and share locations to customers.	Quick link that bundles the locations into the app.

Market need: Consistent flow of repeat customers.

Stakeholder	Challenge	Solution
POI owners	Clients don't like their social media feed being filled with companies.	Follow POI is done via the app, updating users on offers for places.

2.5 Design opportunity statement

Navigation today is aimed towards efficiency, we have lost our joy of exploration. Society is shifting towards an increasingly online model causing brick-and-mortar shops to fail and lowering overall wellbeing. Two Step aims to bridge the online world and bring back our joy through discovering hidden gems in your local area, helping users reconnect with local businesses.

3. Design Journey

After defining the design opportunity, we embarked on a design journey to ensure TwoStep fulfilled our user and partner requirements. This section outlines the primary and secondary research conducted to identify the relevant requirement which were refined through user interviews and validation testing.

3.1 Initial User and Desk Research

Our initial phase involved conducting desk research into user views on current travel apps, and conducting interviews with potential users to determine an initial list of requirements which resulted in the following insights:

Initial User Insights

1. Users feel they have "*lost the joy of exploration*" as navigation apps are geared towards efficiency.
2. The efficiency and safety features of current apps is still highly valued.
3. User expressed an interested in personalised recommendations that were tailored to their specific interests: "hidden gems", historical sites, local cuisines or scenic paths.
4. Users showed interested in recommendations from locals or popular spots suggested by other travellers.
5. Users want the ability to spontaneously start walks, but also wanted the option to adjust the suggested structured itinerary thus allowing for flexibility.
6. Users want the option to save their paths or routes to avoid connectivity issues whilst out.

Initial Desk Research Insights

1. Growing trend for experiential travel, users are more focused on memorable experiences [29]
2. 76% of people want more sustainable transport options but cost is a limiting factor [30]
3. Greater appreciation the community created when sharing recommendations online [31]

3.2 General Survey Results

The second stage of design research focused on understanding consumer behaviours and analysing emerging trends. A general survey was shared with our potential user groups across online communities and forums, our key findings were as follows:

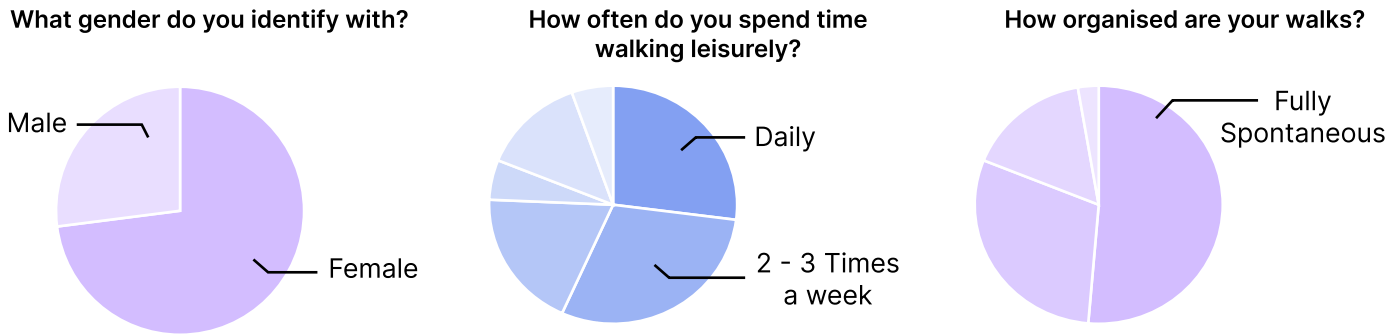


Figure 10 - Survey results overview (section 1)

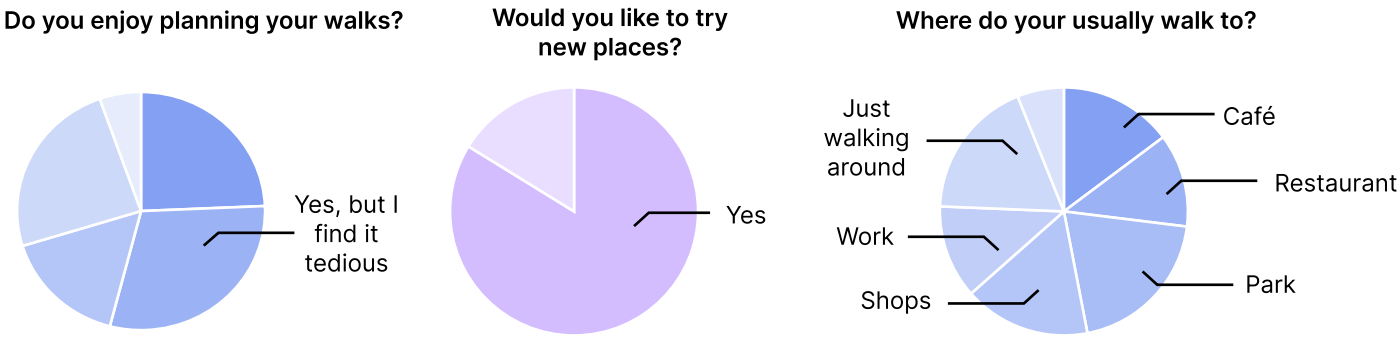


Figure 11 - Survey results overview (section 2)

3.3 Design Specification

1. Provide a spontaneous walking route that users can optionally customise with ease.
2. Allow users to prioritise specific features of their walk; for instance, scenery or passing by a cafe.
3. Provide users with 'to go' routes that don't need planning.
4. Provide personalised recommendations of local POI to visit; including cafes, parks, restaurants.
5. Provide users with the option to record and share their walking routes with friends.
6. Allow users to save their routes offline.

3.4 Developing Personas

Based on comprehensive analysis of our initial interviews, research and survey results, 2 personas were developed to better understand the needs and motivations of our users, ensuring Two Step catered to their specific requirements.

Enthusiastic Erica

Spontaneous Walker

Age: 30

*Erica has been focusing on her **health and wellbeing** and likes to go on **scenic spontaneous walks** a couple of times a week, but doesn't have time to plan walks with her working schedule. She also likes to visit local coffee shops on her walk.*

Requirements:

- She doesn't have time to plan specific routes and wants a walking route tailored to her needs.
- Be able to save routes she likes.
- Wants incentives to go on walks.

Curious Carlos

Tourist

Age: 27

*Carlos loves **travelling** and exploring new cities. He likes to wander the city and go to food places or attractions recommended by **locals** but struggles to find these places without researching about the city.*

Requirements:

- Wants to focus on the experience of exploring the new city.
- Wants local recommendations added to his routes.
- Doesn't want to have to research the city too much for walks.
- Customisation of the 'spontaneous' route to include places he knows he wants to visit.

Figure 12 - Persona generation

3.5 Concept Ideation

Various solutions matching our market space were created and ranked accordingly against feasibility and how well they fulfilled our user requirements. The three final concepts were selected as such:

1. Safe Navigation App

A navigation app which prioritises the safety of the route, encouraging users to reroute the directions away from potentially dangerous areas. Users are be able to communicate to others the level of lighting along streets as well as how busy roads are so that others new to the area can have a safe walking route.

2. Customised Navigation App

A navigation app that generates a random walking route for users to explore new places. Users have the option to customise their route and prioritise the experience of the walk over efficiency. Users can also input specific food spots or attractions they want to visit in their walk.

3. Experience Navigation App

Instead of the phone screens demanding constant attention to deliver directions for the route, the phone vibrates to communicate the live instructions to the user. This allows users to focus on the area that they are exploring rather than be distracted by devices.

3.6 Concept Evaluation

Based on our defined design specification, our 3 concepts were evaluated and Customised navigation was taken fowards.

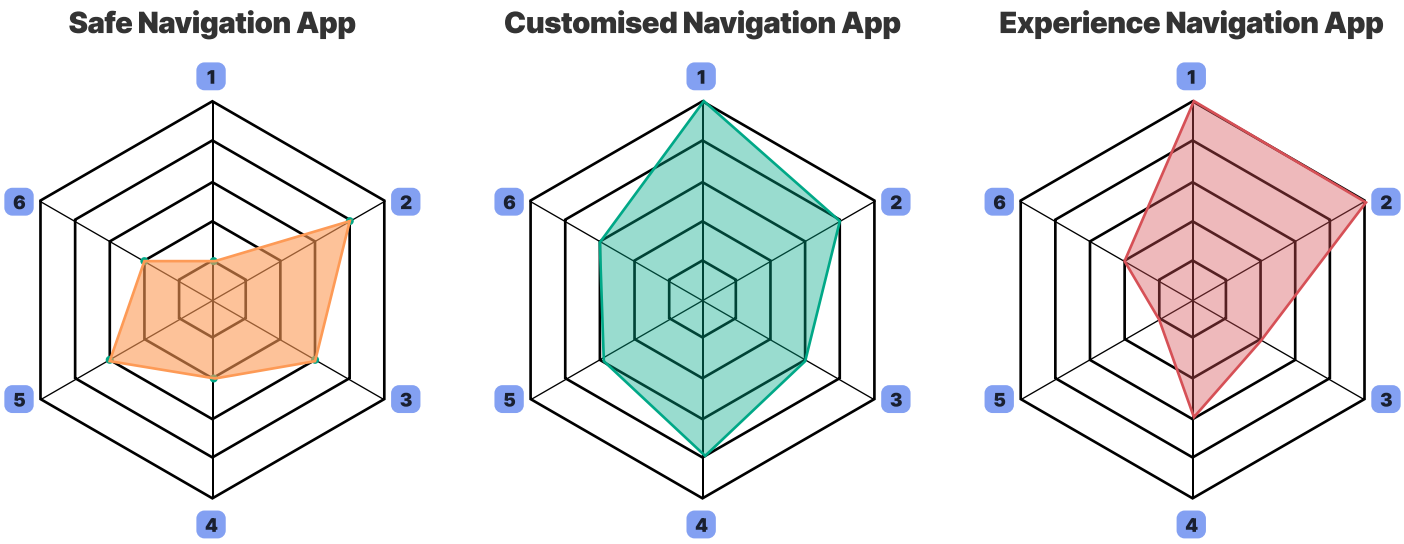


Figure 13 - Concept Evaluation

3.7 Initial UI/UX Prototype

An initial prototype was developed to demonstrate the concept for further user testing.

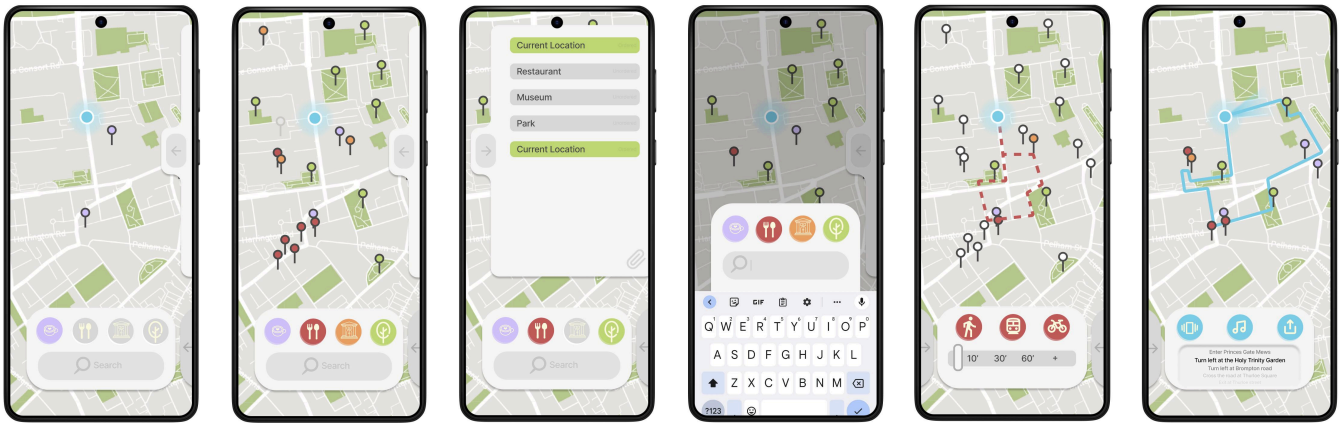


Figure 14 - Demonstrating the UI/UX Interface

3.8 Park interviews

Two Step understands success relies on it's adoption and appreciation by it's users. To that extent, additional interviews and user testing were conducted to to refine key app features and user experience. These interactions were pivotal in refining our prototype and overwhelming positive feedback proved there is sufficient demand for this app:

All interviews were conducted with **spontaneous walkers** validating our core users.

Interviews were conducted with 5 groups (12 people in total)

10/12

Responded positively to the concept

Figure 15 - Participant breakdown

Key Insights

1. Users want the ability to record and share their routes
2. Users want the number of steps included
3. Reward system to incentivise walking outside
4. Include pubs in personalised recommendations
5. Dietary requirements option for recommendations

3.9 UI/UX Prototype Iteration

Detailed stats and essential gamification were added to the core features:

Figure 16 - Demonstrated the further developed UI/UX Interface

3.10 Partner Research

Providing personalised recommendations becomes much more meaningful when partnering with local businesses. As such, the marketing budgets and plans of small businesses were analysed [3] to understand how best to meet their needs within our in app advertisement opportunities.

63%

Of businesses have already **increased** their **digital marketing budgets**

72%

Of overall marketing budgets get put towards **digital marketing channels**

5-10%

Of local business's revenue goes towards **digital marketing**

£700-783

Is spent by small to medium sized businesses for **PPC campaigns** [32]

3

Consumer purchases are needed to create brand loyalty which can be a **struggle** for many small businesses [33]

129,000

In store visits generated by using **location targeted advertising** for No kid Hungry, raising $1 million for charity [34]

3.11 Small Business Interviews

Interviews were conducted with small businesses to gather feedback on the idea from a partner perspective and better understand their unique requirements. This influences how we incorporate advertising into the UI/ UX.

Interview 1 - Café

*"Being in a tourist town we **rely on people walking past** and coming into our shops. It's is something our business would be **happy to put lots of money into every year**. I think lots of other businesses owners on the area would want to be involved too."*

Tim Hayward, Co-owner of Fitzbillies

Interview 2 - Café

*"I think it'd be more beneficial for **places slightly outside of city centres** because if lots of places in the centre were on this app it would become slightly redundant"*
"This seems like a cool idea genuinely, I would need to be more certain on pricing before committing but I would be keen"

Percy Whyley, Manager at FourtyFive Vinyl Café

Figure 17 - Illustrates the key quotes from our interview with cafes

Interview 3 - Escape room

"*We are located on a busy road so often get* ***foot traffic****, however many of my friends* ***struggle to get people in the door*** *so I think this would be really useful to them*"

Harry Janes, Owner of Lockhouse Escape Games

Figure 18 - Illustrates the key quotes from our interview with an escape room

Key Stakeholder Insights

1. More clarity regarding the costs of the advertisement, would prefer a subscription model to be featured.
2. Especially useful for businesses outside of the city centre
3. Smaller businesses rely on flyers, word of mouth or busy pedestrian areas to gain customers

In response to these insights we made the following changes:

1. Instead of partners choosing a price per click, they provide their advertising budget and we determine the respective CPC.
2. Repeat recommendations provided if a user likes a specific place.

3.12 UI/UX Advertising

Based on the insights from the stakeholder, we incorporated a promotional element to the UI, which would be used to promote certain businesses in the area to users, which local businesses would pay for.

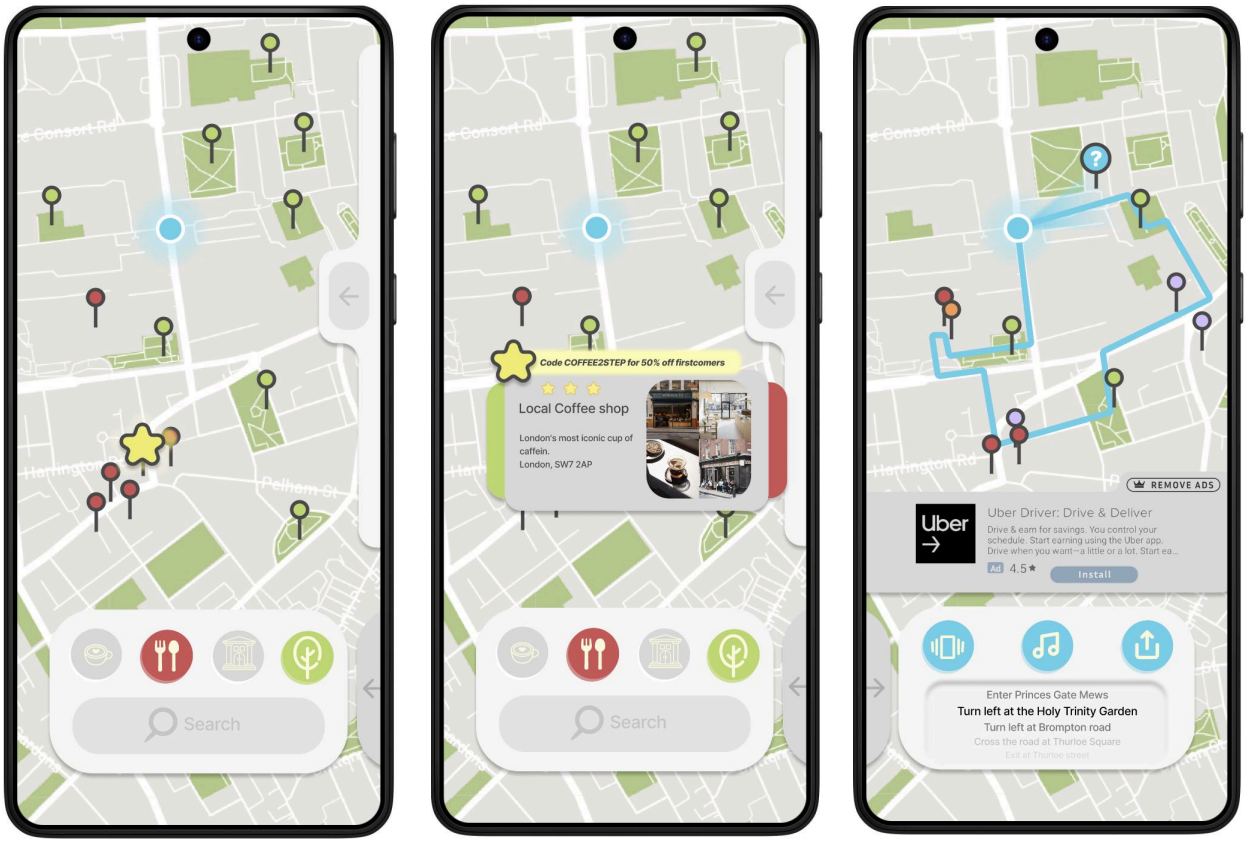


Figure 19 - Shows how advertising has been incorporated into the UX/UI

4 Market Opportunity Calculations

4.1 Top Down Approach

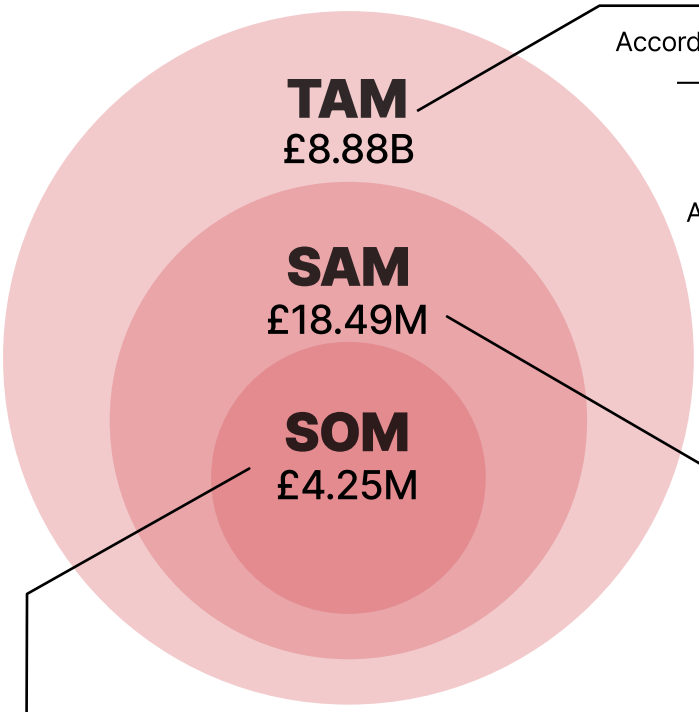


Figure 20 - Top Down Analysis

1 TOTAL ADDRESSABLE MARKET

According to **Statista**, total revenue in the navigation app market — *which covers apps that give real-time travel directions or enable the hiring of transportation services* — reached $1.62B USD ≈ **£1.27B GBP in 2024**. [8]

Another study, conducted by **Business of Apps**, predicts a total revenue of $21B USD ≈ **£16.49B GBP in 2024**. [35]

Since these two statistics differ widely, an **average of £8.88B** was taken.

2 SERVICEABLE AVAILABLE MARKET

Within the navigation app market, the **UK is ranked third** globally in terms of revenue, accounting for $133 million USD (≈ **£104.47 million GBP**) in 2024. [8]

In the first few years, we aim to focus solely on **London**, before later expanding to different cities.

Currently, navigation apps like Google Maps, Apple Maps, Citymapper and Waze reach around **35% of internet users in London** a day, compared to only **26% of the internet population across the rest of the UK**. [36]

There are **6.77 million** internet users in London, and **42.29 million** in the rest of the UK. [37]

35% of 6.77 million ≈ 2.37 million
26% of 42.29 million ≈ 11 million
} 2.37 + 11 = 13.37 →
2.37 ÷ 13.37 ≈ 0.177 → **17.7%**
11 ÷ 13.37 ≈ 0.823 → **82.3%**

So, London accounts for about 17.7% of total navigation app users in the UK. From this we can approximate that the navigation app market revenue in London in 2024 is 17.7% of £104.47M, which is **£18.49M GBP**.

3 SERVICEABLE OBTAINABLE MARKET

The competition factor in this case is hard to determine exactly as consumers often use multiple navigation apps, depending on their needs at that given moment.

Google Maps holds 67% of the market, the rest of the market is divided between more niche apps (e.g. Komoot, AllTrails, Waze, etc.), and represents users whose needs are not currently met by any major player. [38]

As a new and niche startup, we can assume we might capture only **5%** of the 67% held by Google Maps as it is an established, mainstream competitor. We can however hope to capture between **15-20%** of the remaining market shares as it is very divided, indicating that no one app is adequately satisfying user needs.

Taking the average value, we obtain an estimated **competition factor of 23%**, meaning our serviceable obtainable market size is **£4.25M**.

4.2 Bottom Up Approach

In contrast to the top down approach, the bottom up approach provides a more comprehensive overview of the market by analysing the individual customer segments and the revenue acquired from each one.

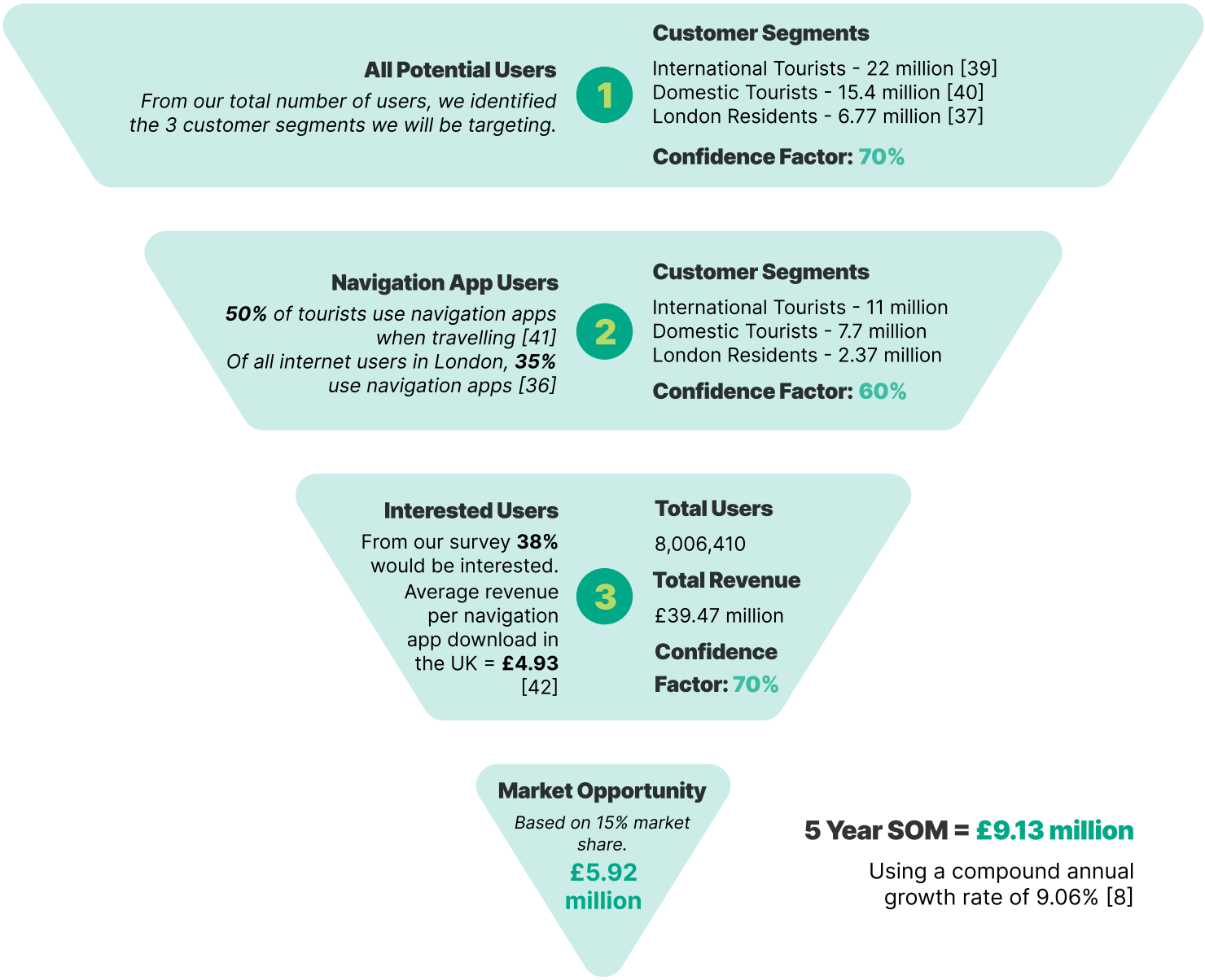


Figure 21 - Bottom Up Analysis

4.3 Value Theory Approach

A value theory approach was used to establish how much users would pay for the service. The value drivers were mapped out and the lifetime value was calculated to provide an Annual Revenue Per User (ARPU) of **£39.01**.

Experience Focused
User's can generate a journey focused on the walking experience rather than the destination

Personalised Recommendations
Users gain personalised food, drink or attraction recommendations

Health and Wellbeing
The walks have a focus on the health and wellbeing of the user.

Spontaneity
Users can generate a random walk without needing a specific destination.

Sense of Community
Provides users with the ability to plan walks with friends or family.

New Area Exploration
Users (particularly tourists) don't need to research the place to explore it.

Yearly Market Opportunity using ARPUR = £46.8 million

4.4 Assumptions and Sensitivity Analysis

4.4.1 Assumptions

Customer Segment Assumptions

For our bottom up analysis predictions were made for the customer segment sizes, based on acquired data.

International Tourists →

- Prior to 2020, the number of international tourists stayed consistent around 21 million. and now this number is returning to pre-pandemic levels
- There is evidence that the number is increasing steadily
- Thus a prediction of **22 million** was made

Domestic Tourists →

- Prior to 2020, the number of domestic tourists stayed constant around 11 million, but jumped to 15.4 million during 2022
- Due to the data behaviour since, we predict the number will now stay constant
- Thus a prediction of **15.4 million** was made

Customer Segment

38% →

- Assumptions about the users interested in the app were made from our survey [section 3.2] and resulted in a value of 38%

Customer Revenue

£1.66 →

- Accounts for IAPs, one-time purchases within the app and in-app advertising
- The same value was applied to all app customer segments despite domestic tourists and London residents having significantly higher app usage than international tourists as overall revenue from each segment will balance out

Market Expansion

9.06% →

- A compound annual growth rate of **9.06%** was used as found through market analysis [8]
- Assumes consistent and sustainable growth of the market
- Any changes that affect this will have an impact on the entire market size estimation

Market Scope

For the top down and bottom analysis two differing competitor factors were used.

Top-Down: 23% →

- A competition factor of 23% was used as a rough estimate as this method exclusively looked at navigation app users in London. We felt we could capture 5% of the 67% share Google Maps holds, and a bigger percentage (15-20%) of the rest of the market as there was no standout competitor.

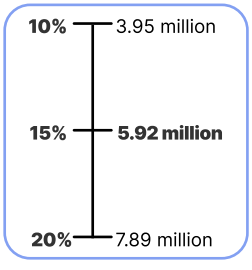
Bottom-Up: 15% →

- A competition factor of **15%** was used as this method considered exclusively interested users.
- Therefore there is less competition for our app and we will be able to access more of the market.

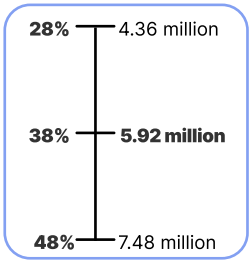
4.4.2 Sensitivity Analysis

Specific assumptions were taken forward to analyse their impact on the calculations for the bottom up should they change, and to assess if they needed to be checked for validity, this was done for only the bottom-up analysis as it is more representative of the users we would get:

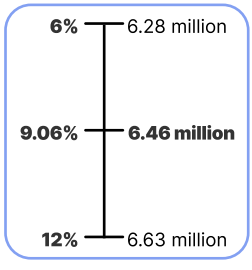
The projected 15% of the market share will vary.



The percentage of interested users may vary.



The projected CAGR of 9.06% will vary.



Potential Bias and Limitations:

- The accuracy of the survey could introduce biases, assuming it represents a broader walking population and overlooks demographic specific biases.
- There could also be innovations by current or new competitors that disrupt the market share.

Projected market share will have the greatest affect on the market size calculation.

5. Business Model

At this stage, insights from market research, analysis, and identified opportunities were combined into a provisional business model. This framework allowed for further refinement and development through ongoing assumption testing, scenario analysis, and financial planning. All components of the model were carefully evaluated to ensure its robustness and potential for success.

5.1 Business Model Canvas

Key Partners

- Local businesses
- Local tourism centres
- Navigation data providers
- Local attractions
- Marketing agency

Key Activities

- Route planning algorithm
- App maintenance, updating GPS data
- Partnering & marketing

Customer Relationships

- Discounts for first visits to new locations
- Gamification
- Feedback forum
- Building community

Channels

- App stores
- Targeted social media adverts
- Influencer partnerships
- Outdoor ads
- Organic sharing
- Online website ads

Key Resources

Technological

- GPS mapping data
- Local business data
- Business review data

Human

- App development team
- Marketing and Advertising team

Value Proposition

Users

- Interest focused navigation
- Multi destination
- Route sharing
- Spontaneous and simple journey planning
- Customisable
- Mindfulness opportunity

Sponsors

- Gaining consumers (ADs)
- build loyal customer base

Customer Segments

- Walkers
- Active friend groups
- New residents of a city
- International tourists
- Domestic Tourist
- Families

Cost Structure

Operational

- Employee salaries
- Pension schemes
- National Insurance
- Recruitment
- Access to mapping data
- App hosting service
- Office hire
- Office facilities and bills

Marketing

- Social media sponsored posts
- Influencer partnered posts
- Outdoor ads
- Online ads

Revenue Streams

Freemium model

- No advertisements
- Further gamification access

Affiliate Partnerships

- Earning commissions by partnering with local businesses benefiting from foot traffic generated by the app
- Focused on restaurants, cafes and shops along the routes

Advertisements

- Earning commissions from amounts of clicks users

Figure 22 - Two Step's Business Model Canvas

5.2 Surface Level System Map

All available routes can be accessed from our app. Walkers can choose to buy the membership for an ad free experience where they can customize and select unique routes, or use Two Steps free model where companies' ads are displayed. Partnerships with business ensure their businesses are featured along suggested routes in exchange for monthly subscriptions.

1. Partnerships with Businesses Revenue Stream
2. Ads Revenue Stream
3. Subscriptions Revenue Stream
4. Businesses Exposure

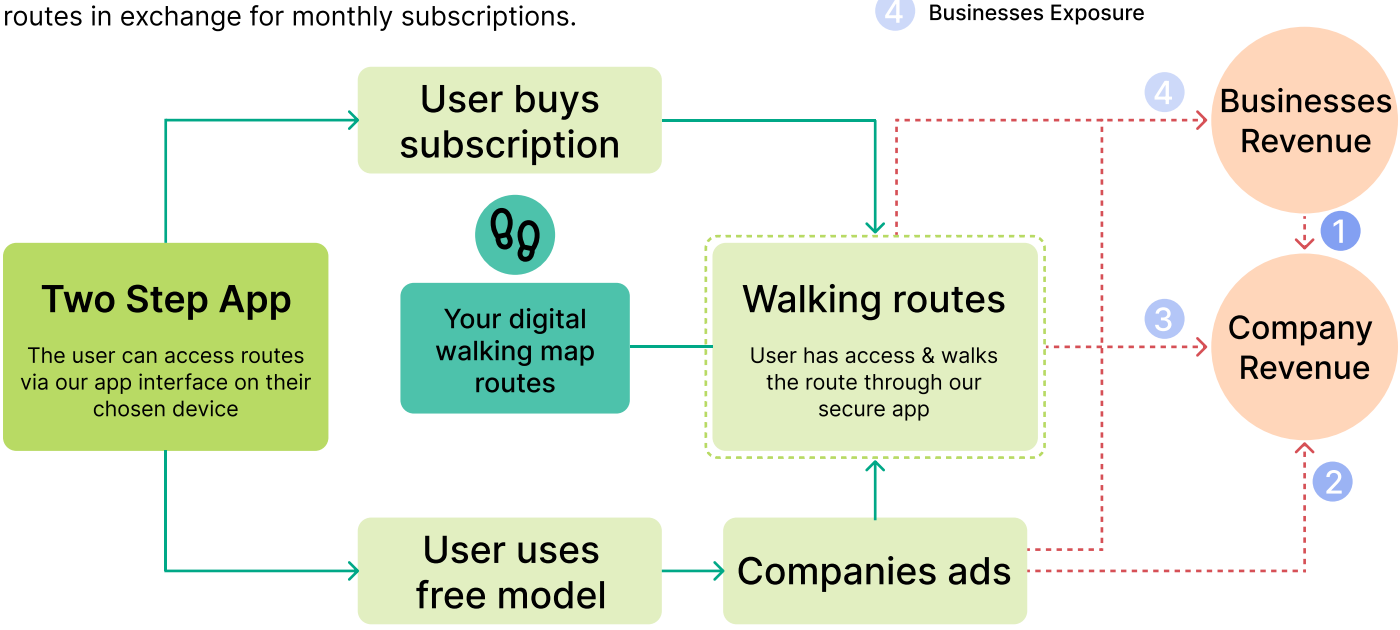


Figure 23 - Two Step's Simplified Revenue Stream System Map

5.3 Two Step's Revenue Stream

1. **Partnerships with Businesses Revenue Stream:**
   Partner with local businesses (cafes, shops, galleries, etc.) along suggested routes earning £0.5 per click
2. **Ads Revenue Stream:**
   Businesses feature their products/services earning £0.3 per click
3. **Subscriptions Revenue Stream:**
   Free Tier: Includes sponsored recommendation and banner ads for local businesses/events.
   Premium Tier (£2.49/month or £29.88/year): add-free experience
4. **Businesses Exposure**
   Thanks to ads and suggestions during walks business get exposure and increased foot traffic

Total Revenue

5.4 Cost Structure

We the broke down Tow Step's cost structure and compared it to that of Citymapper, one of our biggest wayfinding competitors in cities. Although Citymapper is generally used for urban transport, especially public transport instead of non-efficient focused wayfinding routes, it was chosen for their similar app and revenue structure.

	Price	Two Step's%
Partnership Fee	£0.50	100%
Ads paperclick	£0.30	100%
Subscriptions Fee	£2.49	100%

Table 1 - Two Step's Cost Structure

	Price	Citymapper's%
Partnership Fee	-	-
	£0.50	100%
Subscriptions Fee	£1.49	100%

Table 2 - Citymapper's Cost Structure

6 Value Proposition

6.1 Users Value Proposition Canvas

Two Step understands that it's success relies on it's adoption and appreciation by it's users. The figure below demonstrates that the users will find the app desirable.

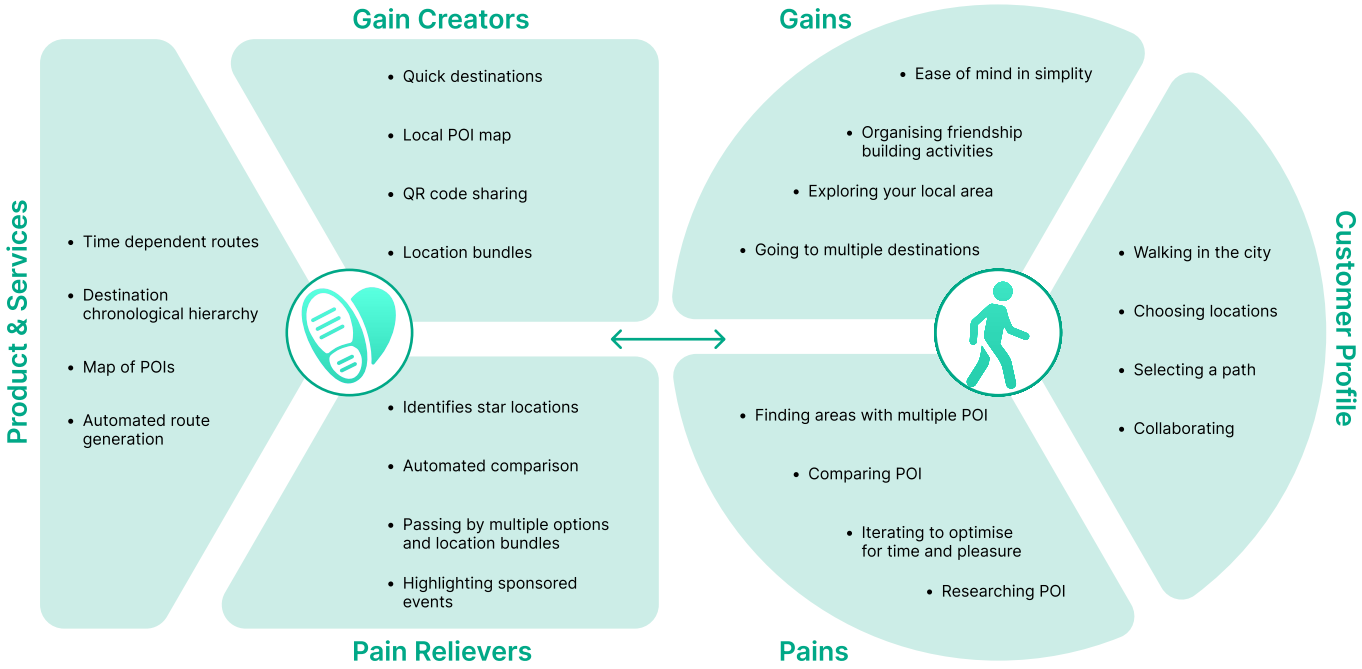


Figure 24 - Value Proposition Canvas for Users

6.2 Partner Value Proposition Canvas

Our platform provides new ways of reaching the target audience of stores, attractions, and other points of interests. Insuring that our partners are satisfied is crucial to insure the financial success of the app.

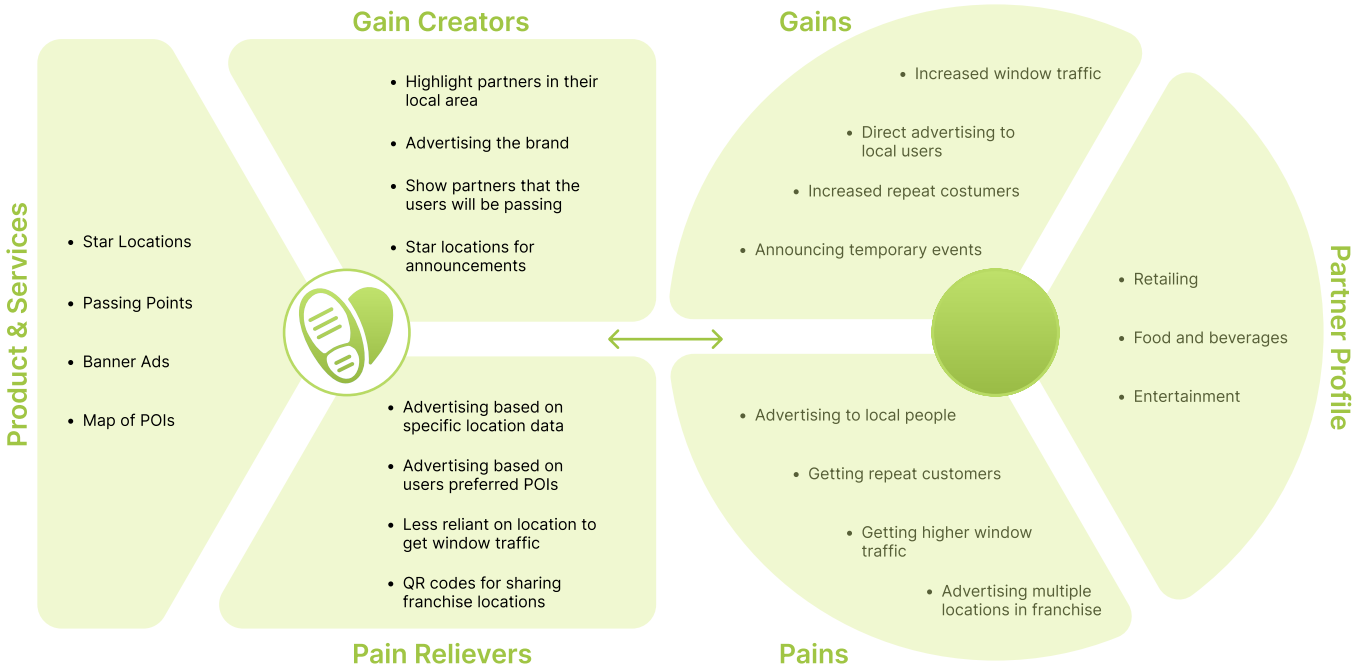


Figure 25 - Value Proposition Canvas for Two Step's Partners

6.3 Customer Value Pricing

APPs	Local POIs	Multi-destination	Public transport	POI ratings	Audio navigation	Route Sharing	Gamification	Round Trip	Local events	User content	Weather	Avoiding Areas	Traffic data	Offline
Two Step	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗
Google Maps	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗
City Mapper	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗
AllTrails	$	✗	✗	✗	$	$	✗	✓	✗	✓	$	✗	✗	$
Komoot	✗	$	✗	✗	$	$	✗	✓	✗	✓	✓	✗	✗	$
Pokémon Go	✓	✗	✗	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
GooseChase	✗	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗
Roadtrippers	✓	✗	✗	✓	✓	$	✗	✓	✗	✓	✓	$	$	$
Yonder	✓	✗	✗	✓	✗	✓	✓	✗	✗	✓	✗	✗	✗	✗
Truffel	✓	✗	✗	✓	✗	✓	✗	✗	✓	✓	✗	✗	✗	✗

Figure 26 - Ranked Competitor Feature Table

Figure 26 shows a list of features available on apps within our category of leisurely walking, ranking them based on how our user persona would value them. Most platforms are missing key features, with Google Maps being positioned as our primary competitor.

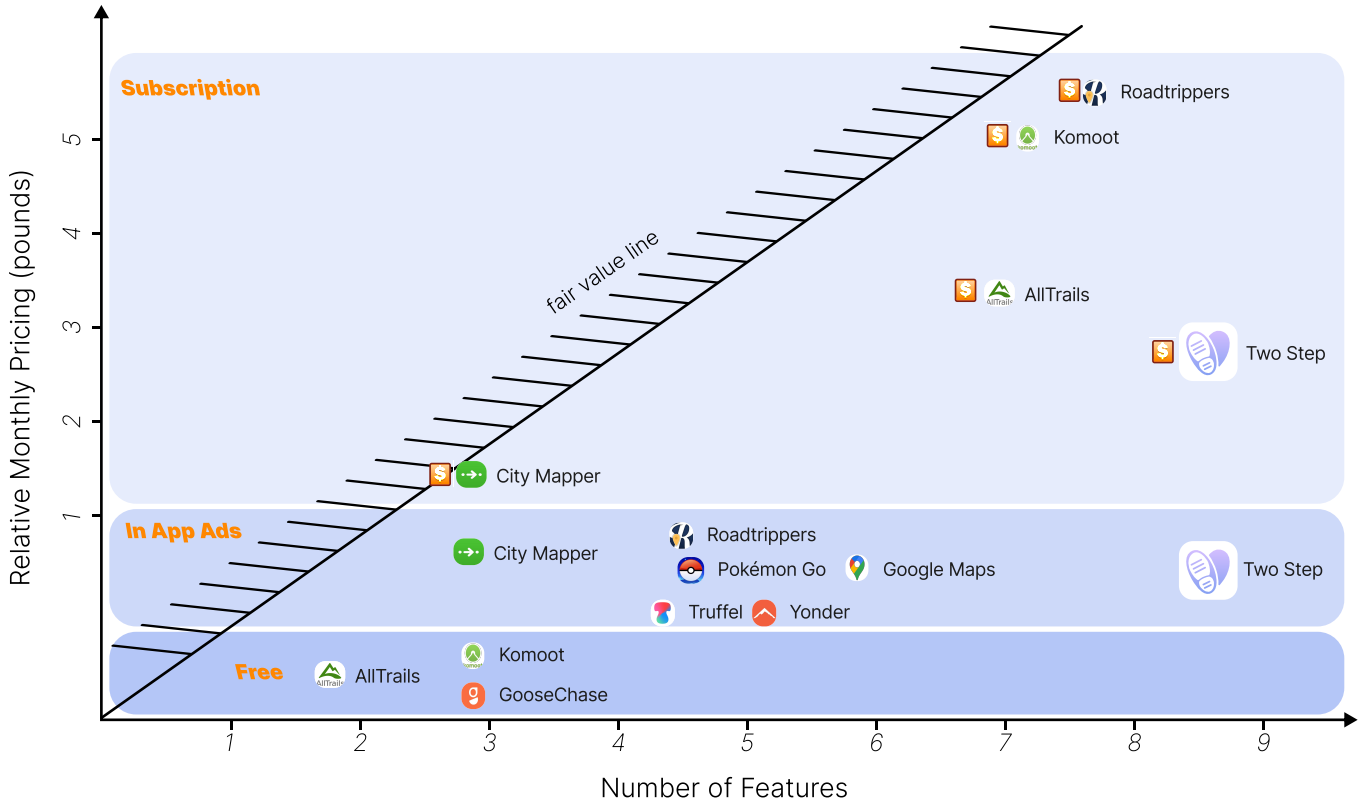


Figure 27 - Customer Value Map (as of 2024)

With the large majority of apps being free to download, or having switched to it (City Mapper), it sets a clear trend as to the pricing of our app. Beyond advertisement, an optional subscription service for our more devoted customers provides additional income. Two Step doesn't wish to lock major features behind a paywall as it would discourage a wider adoption of the app, which is necessary as our primary income comes from our partners. By paying the small subscription fee (2.49 £ / month) users can opt out of receiving advertising and instead receive additional gamification elements.

Two Step's positioning on the bottom right of the value map (Figure 27) ensures that it has more features than similarly priced apps and is less costly than other the high value platforms, placing it at a competitive advantage.

6.4 Partner Value

Having a traditional advertising method which is familiar to advertising agencies and is accessible to a wide range of product, provides revenue from proven advertising methods. These methods often users leaving the app and being redirected to our partners platform, which provides our partners with verifiable metrics on their ad.

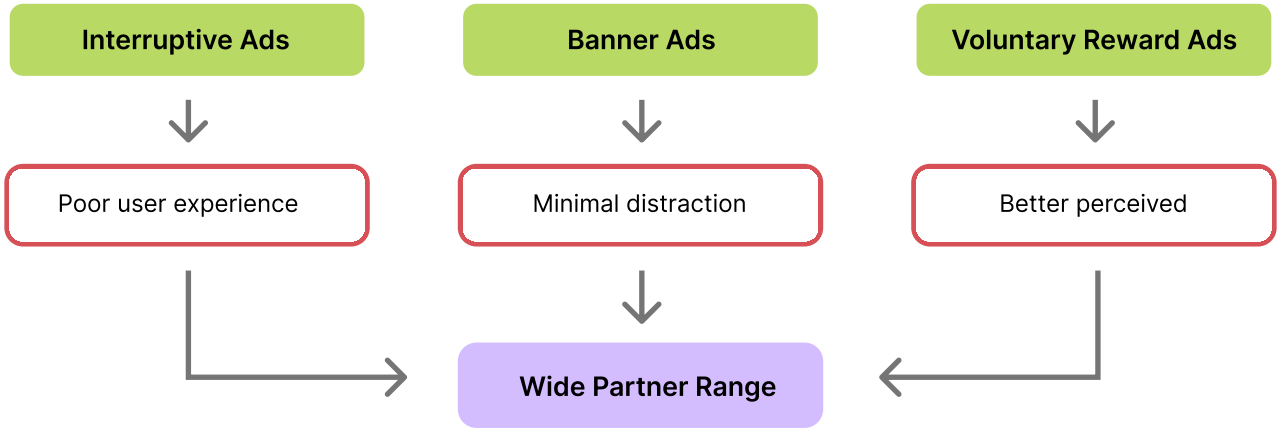


Figure 28 - Traditional General Advertising

A 2022 analysis of app ad review [43] showed that the frequency and quantity of advertising - often concerning pop-up ads (13% of ad reviews), and their interruptive timing (12% of ad reviews) - lead to substantial uninstallations. Banner ads are minimally invasive while providing a proven sales increase for in-person offline shopping [44]. Reward ads tend to be viewed more positively [45] and improve user interaction with the app [46].

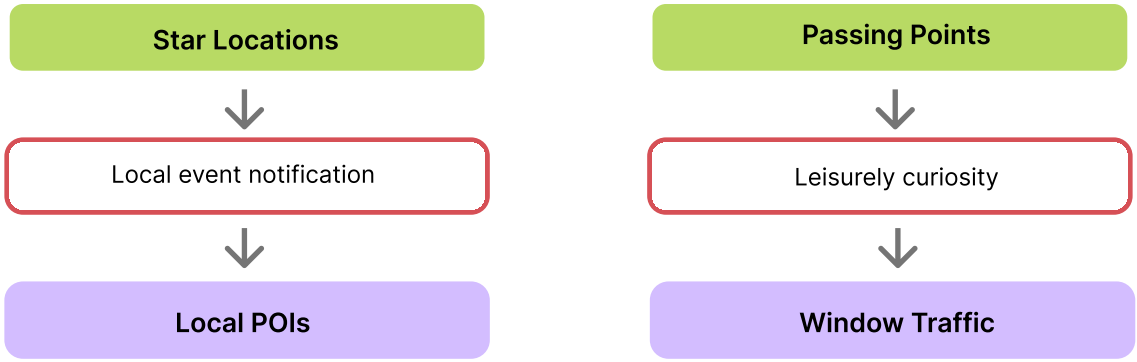


Figure 29 - Integrated Advertising for Points of Interest

Integrated advertising allows for less intrusive ads resulting in a better user experience. This increases the likelihood of users interacting the in-app promotions [47] and thus improves Two Step's revenue. Google uses a system comparable to "Star Locations" [48], and City Mapper has Local Ads which are similar to Two Steps' "Passing Points" [49]. The primary difference is that Two Steps frames these as sponsored content, where users are rewarded for interacting with it. Users positive engagement with the ad determine it's display frequency, with flagging of ads being considered seriously.

For the reasons listed above, our partners will have access to **banner ads** and **integrated voluntary ads.** Banner ads will be placed on the navigation page to capture the users attention - similar to City Mappers content ads [49]. Both Star Locations and Passing Points will be available for our repeat partners only, where advertising costs are be calculated on a cost-per-click model within our partners marketing budget so as to only charge for true customer interaction.

7. Application UI Features

7.1 Simplistic User Journey

Two Step's primary users wish to casually find fun walks. As such, the application's UI was designed to reflect that experience. Two Step is separated into three easy to follow processes which enable a quick, efficient, and clear user journey.

Figure 30 - UI Three Primary Stages

7.2 Control Where it Matters Most

Two Step offers the ability plan journeys with general terms to reflect the import role traveling between destinations has on the choice of a specific location.
Choosing what elements are ordered or unordered allows the users to easily plan circular journeys, or A to B routes with fun stops and coffee breaks.
This process is often more fun when sharing it with a friend, which is why Two Step destination bundles can be shared with just one click. Whether it's to plan a casual date together or collect a weeks worth of possible locations, planning is better as a team.

When on the main map, users can click on specific locations to get a quick summary of what to expect. After which, they can choose to have it added as it's own specific target on the journey, or removed from possible stops. Two Step offers control at every level.

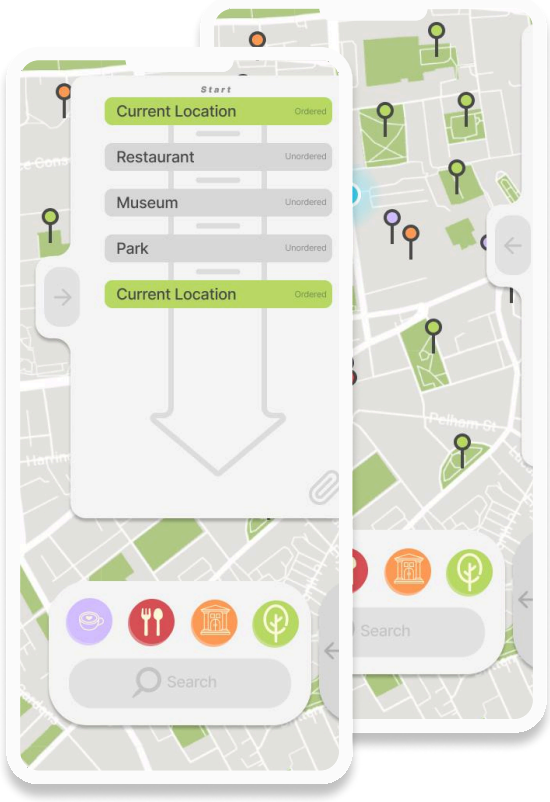


Figure 31 - Destination Control

7.3 Route Generation

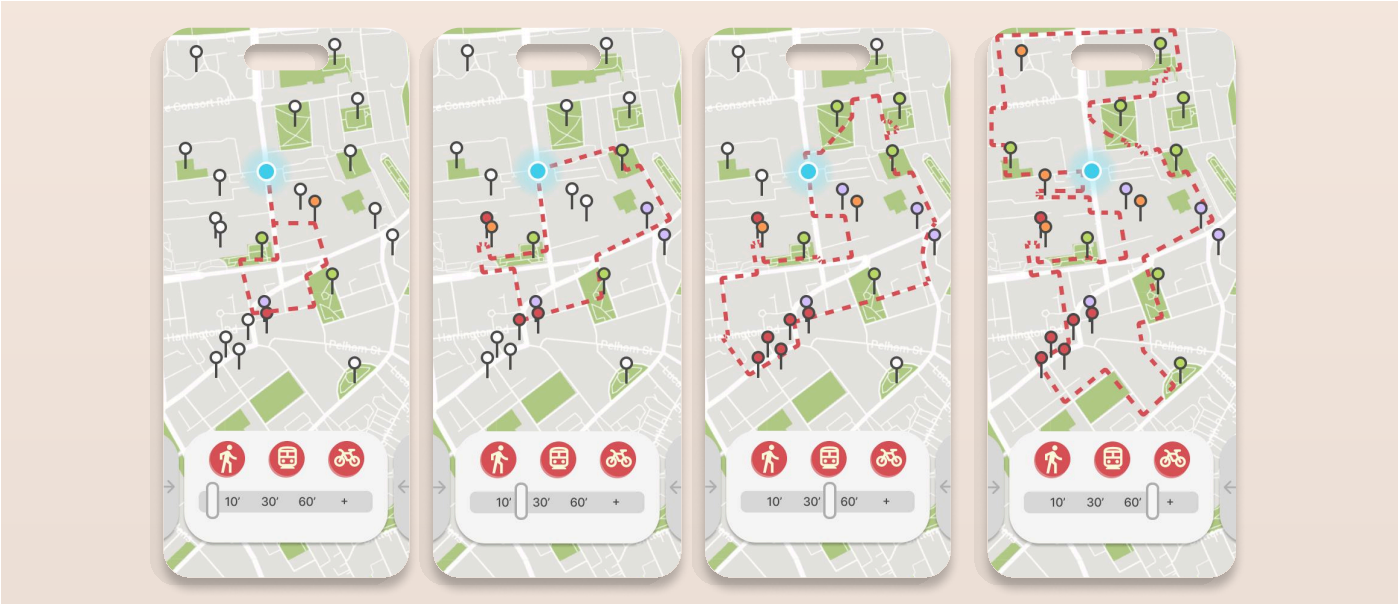


Figure 32 - Dynamic destinations

Visiting an area often evolves weighing time against how much you can visit within that window. With Two Step's dynamic route generation, finding that balance has never been easier. The user can chose a path ranging from going through one of each type to going through every destination.

Figure 33 - Navigating in Two Step

7.4 Navigation UI

Two Step takes a minimal approach to it's navigation UI so as to not be distracting from the traveling experience. We chose to only show a bird's eye view and a single subtitle, providing the user with an idea of where they should be headed with just a glance at their device.
Having vibration cues enables for seamless traveling without disrupting the walking experience. Two Step's audio notifications are designed considering the large volume of users who listen to music while walking. The messages are short and ease in and out with the background music.

The final button displays a large QR code which the user can screenshot or show to others.
When organising a walk within a group, sharing the planned route is crucial for keeping everyone on the same page.
Individuals who plan ahead of time will appreciate being able to save their plans, organise them within their gallery application, and import them back into the app when needed.

Figure 34 - Star Locations

7.5 Advertising Star Locations

Two Step's partners can have their points of interests (POI) highlighted in the main map. Two Step is paid based on the number of users who click on the highlighted location.
When a user clicks on it, they are given the usual short summary and review of the place and a short sentence relating to why the location was highlighted. POIs can use this feature to market discounts or events to people in the area.
Considering the risks of notification fatigue, systems will be in place to limit the number of stars a user will see on their page per day. The map will prioritise locations that don't often have stars and those the individual user has previously liked. This ensures that stars keep their value and novelty factor while enabling smaller POIs to compete with larger organisations.

7.6 Passing Points

Window traffic is crucial for many businesses. Two Step's Passing Points feature enables our partners to translate this into the digital environment. The icon incites the users curiosity and has them thinking "Since I'm passing by, I might as well go see what it is". Our partners can either chose to use their own logo to build brand recognition, or use the default Two Step icons to communicate what they do.

7.7 Banner Ads

Non invasive banner ads are a tried and true method of getting advertising revenue. These banners are placed on the final step of the process so as to not interfere with the user's journey planning experience. Once the path is generated, they are more likely to relax and look at the advert.
For our more devote customers, they can sign up for a subscription service and have these ads removed for a truly distraction free experience.

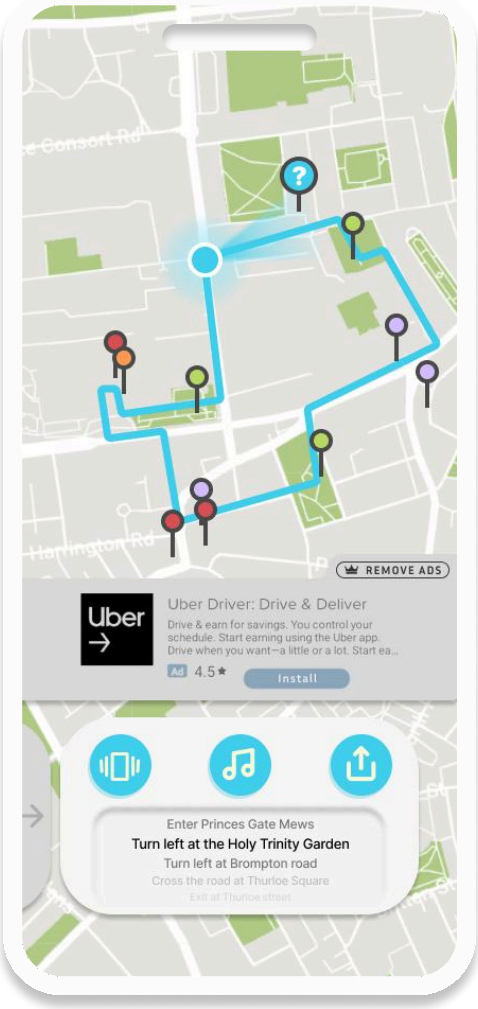


Figure 35 - Banner Ads and Passing Points

7.8 Gamification of Two Step

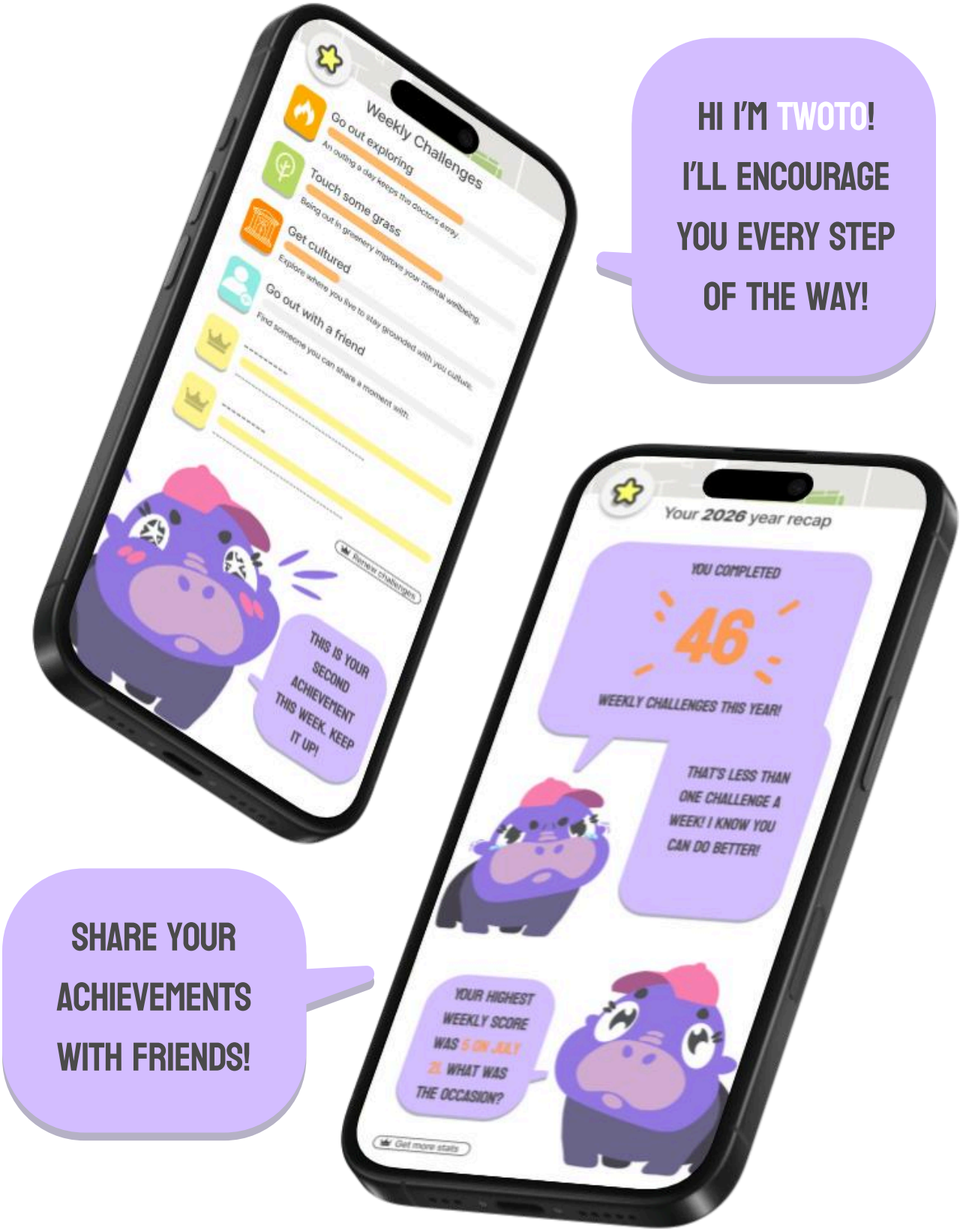


Figure 36 - Twoto is here to help!

Twoto the hippo was inspired by Moo Deng, a baby hippo who gained world-wide recognition and renown through cute videos and photos being shared online [50]. Twoto is at the core of getting users to feel connected to the app, increasing engagement and thus advertising returns. Research has proven that market share of long-term campaigns using a mascot had a gain of 41% compared to 29.7% for those not using a mascot [51]. Yearly reports provide users insight into their habits, and much like Spotify Wrapped [52], sharing these stats helps build the Two Step user community and generate further new users.

8 Assumptions

8.1 Idea Hypothesis Canvas

To define key assumptions, we followed the process outlined by the Imperial Entreprise Lab [53], starting with the Idea Hypothesis Canvas.

PROBLEM

Our theory of the problem:

Current navigation apps are designed to help users get from point A to B quickly and efficiently. However, they lack options for walking leisurely, aiming to discover new places or pass time

The current solutions to this problem are:

Using Google Maps or another map software to pinpoint an area with all destinations desired, picking specific stores/places to visit, then inputting all the different locations into a Google Maps itinerary

Our untested assumptions are:

- Walkers want to be more spontaneous with their walks
- Walkers don't always have a specific destination in mind (a specific restaurant, coffee shop, pub, etc.)

SOLUTION

Our theory of the solution:

A navigation app that generates a randomised route based on the type of locations the user wants to visit (coffee shop, restaurant, etc.) and the time they have.

This solution is unique and compelling because:

- Allows users to plan complex multi-destination routes effortlessly
- Routes are completely customisable
- Allows for round trips without going through the same street twice

Our untested assumptions are:

- Mapping data (e.g., location of open businesses) is accurate and readily available
- Generating randomised yet customised routes is technically feasible
- Phones and apps will continue to be widely used
- In the coming years, there won't be another pandemic or disruption that prevents people from walking leisurely
- The app will be easily downloaded from all major app stores
- The app's UI will be intuitive enough to users for a seamless transition from existing tools
- The app's gamification features will increase user engagement and retention

BEACHHEAD CUSTOMER

Our theory of the beachhead customer:

- Tourists looking to discover the city, don't have go-to spots yet, and seek unique, hidden gems
- Local residents who want to break out of routines and discover new activities or places

The size of the potential market is:

£3.08 million based on the bottom up calculation

Our untested assumptions are:

- Users of all types will want features like curated itineraries and recommended places
- There is a large enough audience that prefers exploration over efficiency
- Users value customisable routes that reflect their preferences and avoid repetitive paths
- Users will find real-time nearby activity recommendations valuable and reliable
- Users trust the app's ability to protect their data

BUSINESS MODEL

Our theory of the business model:

- Freemium model where users can pay to remove ads and unlock certain features
- Most of the revenue comes from ads and partnerships

The price customers will be willing to pay is:

£2.49 subscription fee

Our untested assumptions are:

- Local businesses will pay for exposure or partnerships to attract app users
- Beachhead customers will be willing to pay a small subscription fee for premium features
- Small local businesses in London will stand their ground against corporate chains and continue searching for ways to attract users

8.2 Assumptions Mapping

Having outlined the key assumptions underpinning our app idea, we now need to assess which assumptions present the highest potential risks to the success of our business. In doing so, we can prioritise efforts to validate them, thereby reducing uncertainty and strengthening the foundation of our app.

We first segmented our assumptions into external factors and internal factors to help distinguish between elements we can control (like app features and technical feasibility) and those we must adapt to (like market behavior and external stakeholders), ensuring a more targeted and efficient risk evaluation process.

8.2.1 External Assumptions

External factor assumptions depend on market conditions, user behaviour, or external stakeholders. They are influenced by forces outside our immediate control and it is therefore crucial to validate them as wrong assumptions can lead to misalignment with user needs and market demand. If any of our assumptions are incorrect, we will need to readjust our product and business model.

Rationale (see Appendix C):

Each assumption is scored out of 10 corresponding to **1/ our current degree of uncertainty**, keeping in mind all previous research and validation done, and **2/ the negative impact** it would have on our business if it was false. We made sure to provide justification for each rating. The scores were then multiplied together to obtain a final risk assessment score and identify the top three high-risk assumptions.

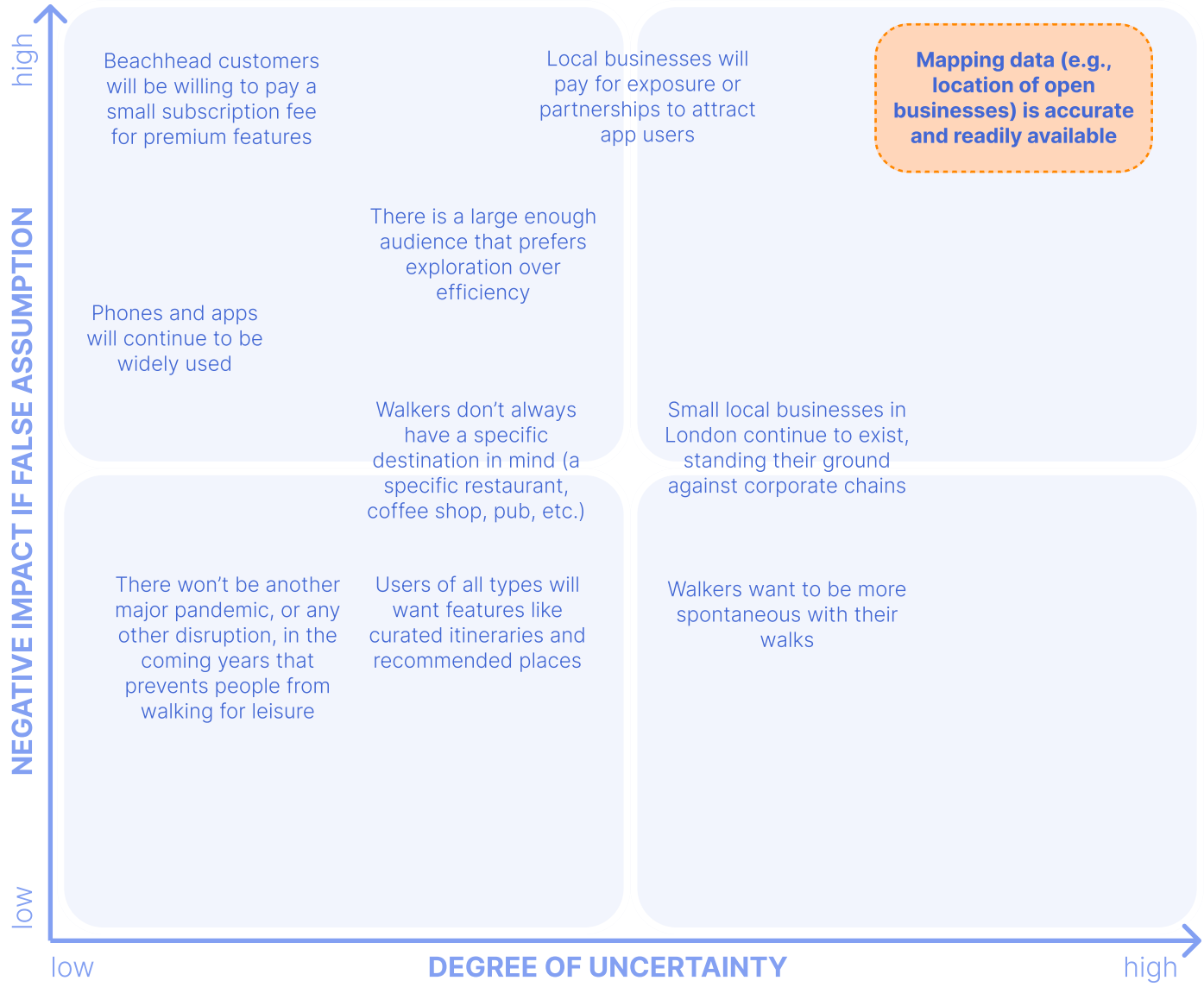


Figure 37 - External Assumptions Risk Evaluation Matrix

8.2.2 Internal Assumptions

Internal factor assumptions are those directly related to elements within our control, such as the app's design, functionality, or technical feasibility. These assumptions are often tied to our team's capacity to execute the project successfully. Addressing these ensures that the app's core functionality, user experience, and development process are realistic and achievable, minimising risks like project delays, resource mismanagement, or unmet user expectations.

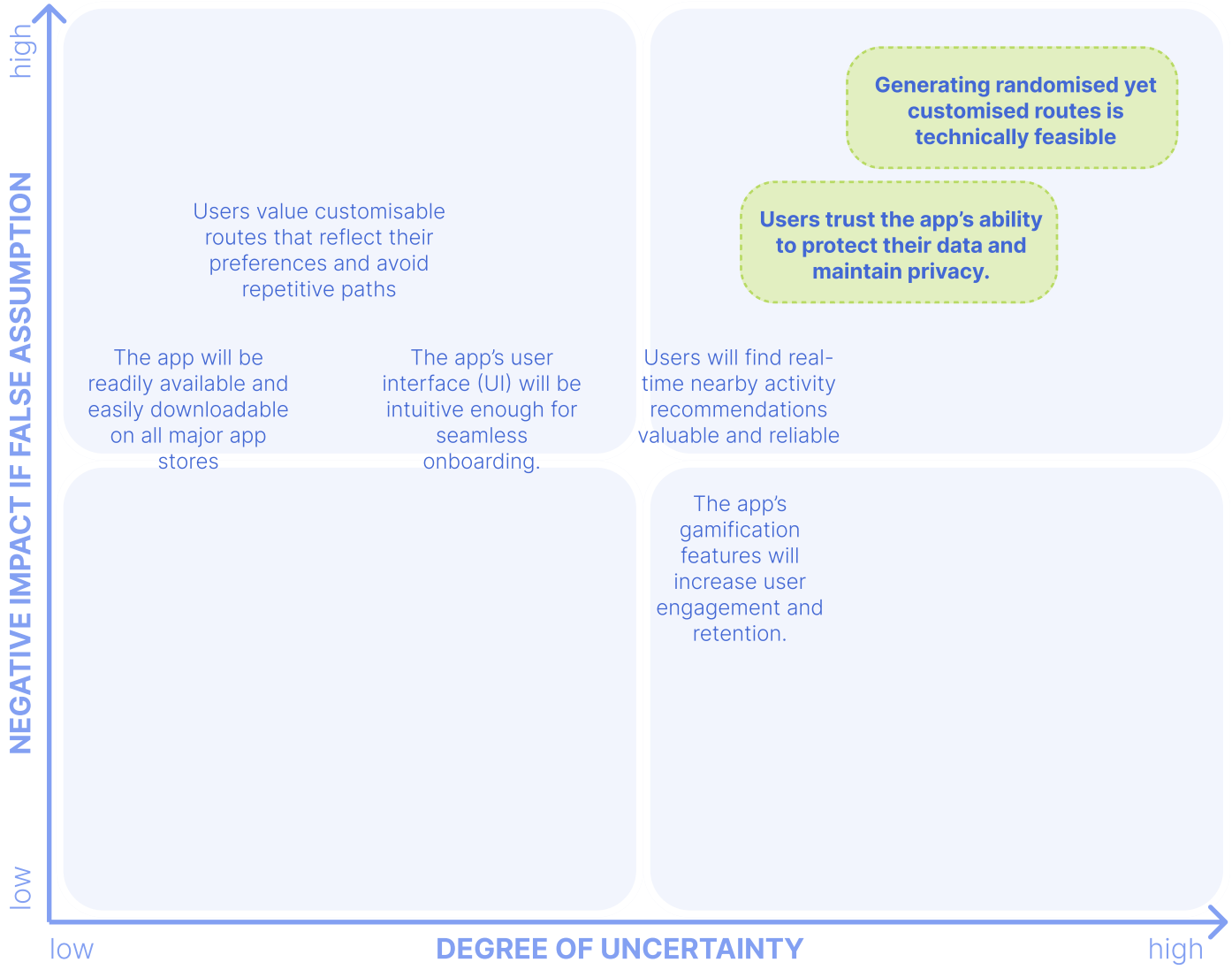


Figure 38 - Internal Assumptions Risk Evaluation Matrix

8.2.3 Justification for the 3 highest-ranked assumptions

Real-time data (e.g., location of open businesses) is accurate and readily available

Two Step relies heavily on real-time data for functionality. For instance, our app requires information about the location of coffee shops, restaurants, etc. as well as customer reviews for each point of interest. Inaccurate or unavailable data would render key features unreliable or non-viable.

Users trust the app's ability to protect their data and maintain privacy.

According to IApp, over the past 12 months, 85% of consumers said they deleted a phone app due to privacy concerns [1]. Additionally, a McKinsey study found that a majority (87%) of consumers say they won't do business with a company if they have concerns about its security practices [2]. It is therefore essential to ensure users trust the app's ability to protect their data.

Generating randomised yet customised routes is technically feasible

Whilst there are many already existing navigation apps, none of them operate quite like Two Step -- with the focus not being on efficiency or time optimisation, and adding an element of randomness. If technical challenges were to arise, they could delay the app's launch or require simplifications. As such, it is crucial to ensure the feasibility of our software.

8.3 Assumption Validation

These assumptions were then validated using different methods in order to mitigate business risks.

1 Users trust the app's ability to protect their data and maintain privacy.

DESK RESEARCH

According to 71% of consumers, they would stop doing business with a company if it gave away sensitive data without permission [54]. 64% also agree that companies that provide clear information about their privacy policies enhance their trust [55].

Based on this, a clear disclaimer was added to the app's terms and conditions pop up when the user first installs the app, indicating that Two Step does not sell users' data (**see Figure X**)

USER VALIDATION

The updated prototype was shown to 8 potential users, who were then asked whether they would have any privacy concerns when using Two Step. 7 out of 8 replied no, whilst one user felt that companies often lied in their disclaimers. Given the small sample size, a larger test would need to be conducted at later stages of the business.

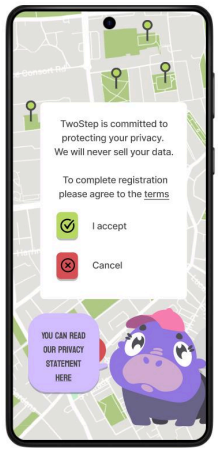


Figure 39 - Disclaimer & Terms and Conditions

2 Mapping data (e.g., location of open businesses) is accurate and readily available

DESK RESEARCH

Many data providers offer structured datasets containing details about businesses, such as location, categories, reviews, and ratings. Several navigation and review websites/apps, including Google Maps, Yelp, and MapBox, also offer APIs which facilitate app development as well as providing access to their extensive databases.

For public transport information, TFL offers real-time data for free [56].

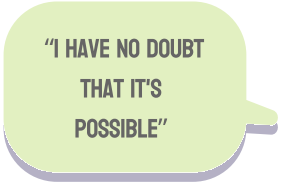
For mapping data — for instance the names of streets and location of businesses, OpenStreetMap offer open license free, editable maps of the world which include: roads, buildings, addresses, shops and businesses, and points of interest [57].

Finally, for user reviews, the Yelp Fusion API stands out as the cheapest option, offering 15 data attributes such as average star rating and closure status for $7.99 ≈ £6.27 per 1000 API requests [58].

3 Generating randomised yet customised routes is technically feasible

EXPERT INTERVIEW: Sasha Walker, Software (DevOps) engineer

We consulted an expert to validate the technical feasibility of our app.



Key Insights:

- She suggested that we estimate walking times using distance heuristics. We should optimise the path and destination choices using trial-and-error with a greedy method, adding the best single stop to the route, and repeating until the route is the correct length.
- She suggested encouraging users to order the stops to reduce the problem space. This is already the default option in our app, which allows us to save computational power.
- Finally, she encouraged us to use the Google Maps API services for the specific navigation instructions.

Overall, Sasha ensured that creating Two Step, though challenging, is feasible.

8.4 Mitigating Critical Business Risks

We analysed the internal and external risks that threatened the business, and developed contingency plans to mitigate them.

Internal Risks

Risk	Impact before mitigation	Likelihood	Mitigation	Impact after mitigation	Likelihood after mitigation
Operational Activities: Investors try to force an unwanted decision or push the business down a different path	MEDIUM	MEDIUM	**Avoid:** Ensure investors and founders are on the same page before starting a partnership	LOW	VERY LOW
Staffing: A founding member leaves	HIGH	MEDIUM	**Active acceptance:** Ensure there are clauses in all contracts to allow for a smooth handover	VERY LOW	MEDIUM
R&D: We can't develop the adequate software for Two Step	VERY HIGH	LOW	**Transfer:** Delegate to a more specialised team of software engineers with more experience	MEDIUM	VERY LOW
Technological: Server crashes or technical glitches	HIGH	HIGH	**Contain & active acceptance:** Regular maintenance, backups, and status checks to reduce the probability of servers overloading. If crashes or glitches still occur, work on repairing them	LOW	LOW

Table 3 - Internal Risks

External Risks

Risk	Impact before mitigation	Likelihood	Mitigation	Impact after mitigation	Likelihood after mitigation
Political: Governments restrict walking outside, like during the COVID-19 quarantine	HIGH	MEDIUM	**Passive acceptance:** Keep developing new features for when the lockdown ends	MEDIUM	MEDIUM
Technological: One or several of the API used goes out of order	VERY HIGH	MEDIUM	**Transfer:** Use an alternative API	MEDIUM	LOW
Competitors: Increased market competition	MEDIUM	HIGH	**Active acceptance:** Conduct regular competitor landscape analyses and adapt to findings in order to keep our competitive advantage and USP	LOW	LOW
Technological: Navigation data hacks impacting user trust and app operation	VERY HIGH	HIGH	**Contain:** Regularly update encryption measures and delegate cybersecurity duties to a specialised tier	LOW	LOW

Table 4 - External Risks

9. Marketing Channels

9.1 Bulls- Eye Marketing Analysis

Due to the diversity of our customer segments, a robust sales strategy is best to maximise Two Step's exposure and engagement with our targeted demographics. Using the 'Bullseye Framework,' the most effective channels for customer acquisition were identified and assessed based on their potential to engage the target audience. Sales channels were evaluated and prioritized, with a focus on those offering the most targeted and impactful reach.

Figure 40 - Bulls-eye Analysis

9.2 Sales Channels

The selected sales channels ensure customer acquisition through app stores ads and physical outdoor ads, which effectively engage pedestrians, tourists, and individuals across various age groups and demographics. Additional channels include future business partnerships, a strong social media presence, organic sales, and word-of-mouth referrals. These channels are expected to grow and evolve over time as the platform builds its customer base and reputation.

Figure 41 - Sales Channels

The following calculates the total marketing costs spent to acquire a 1.2 million userbase. This number corresponding the 15% share of the 8 million potential users shown in the bottom up approach [page 18].

9.3 App Search Ads

We know that people are dissatisfied with current walking navigation apps enough to actively search for alternatives (Survey [1]). Two Step should be promoted when these potential customers search for alternatives. The UK share between IOS and Android mobile phones being evenly split [59], Two Step's search ads budget will be evenly split between them. The apple store is the only permitted store on IOS and Google's Play Store accounts for 90% of app downloads on Android [60].
Google operates on a cost per click model, where a click doesn't guarantee a download [61]. Apple offers a cost per install model within their Basic Plan [62]. The pricing of the ads are dependent on the evaluated quality of the app and the competition within the category. Two Step will start within a niche market and rely on good reviews to boost its ranking paired with an market average cost.

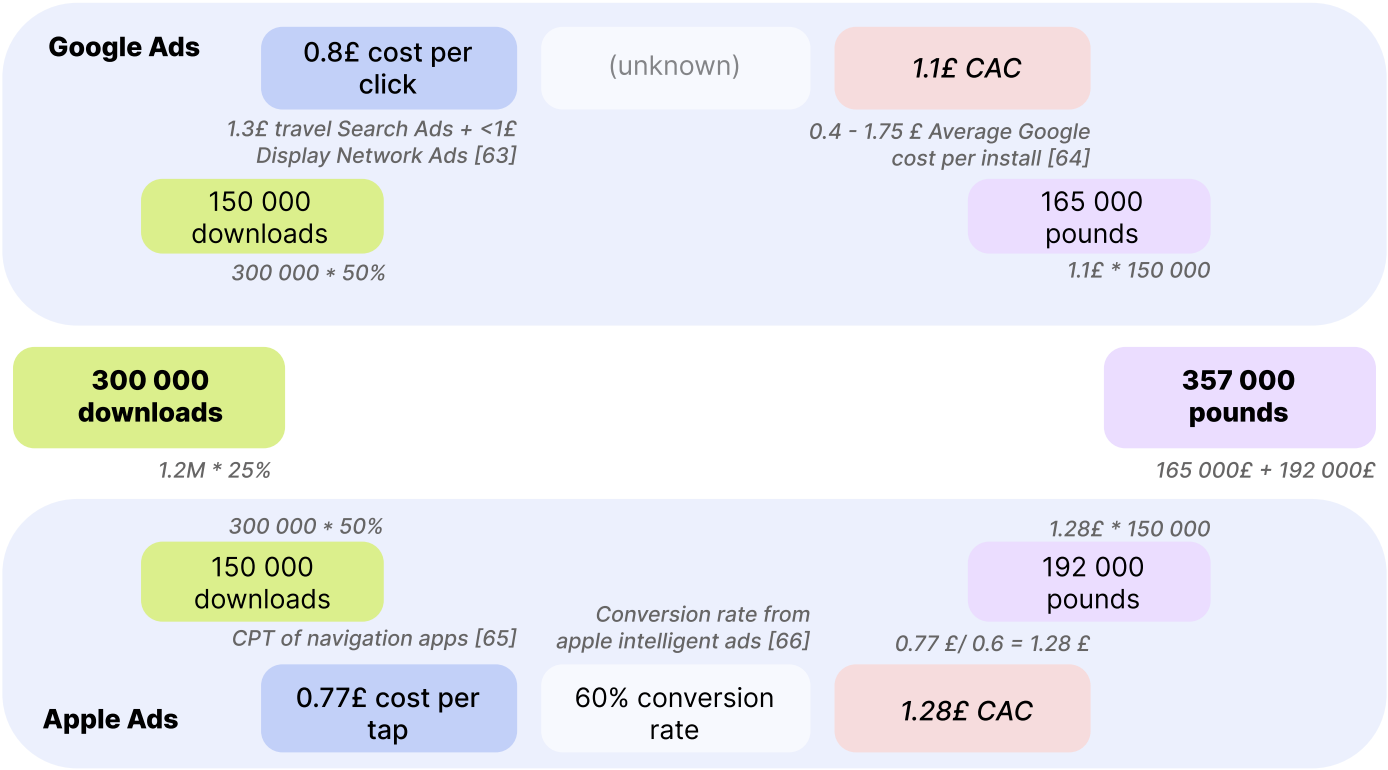
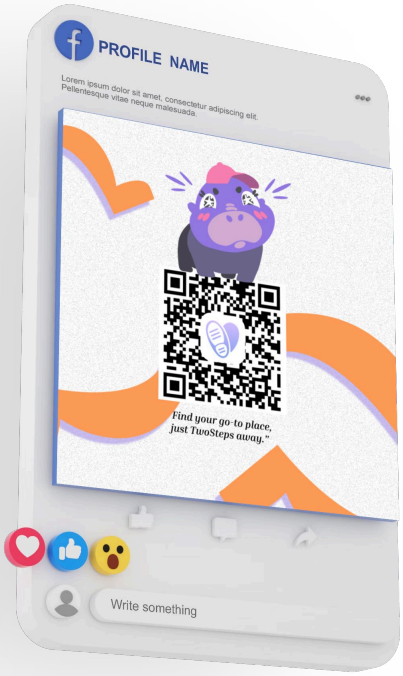


Figure 42 - Total App search advertising costs

9.4 Social Media



80% of UK internet users use Facebook [67] and 60% use Instagram. 30% of our survey responders using social media to find places to walk to. Both of these facts make it clear that Two Step should be advertised with Meta.
Instead of setting a maximum cost per click Meta uses an overall advertising budget system [68].

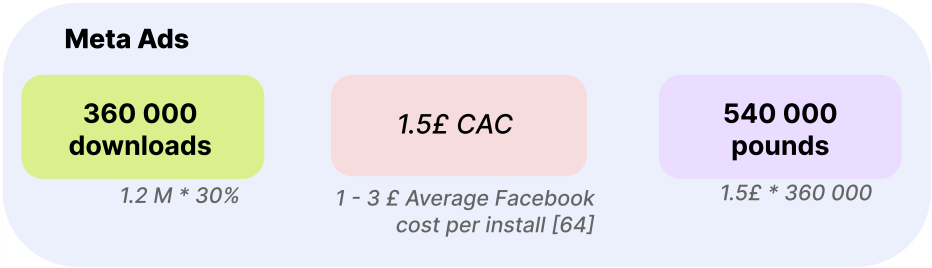


Figure 43 - Total social media advertising costs

9.5 Outdoor Ads

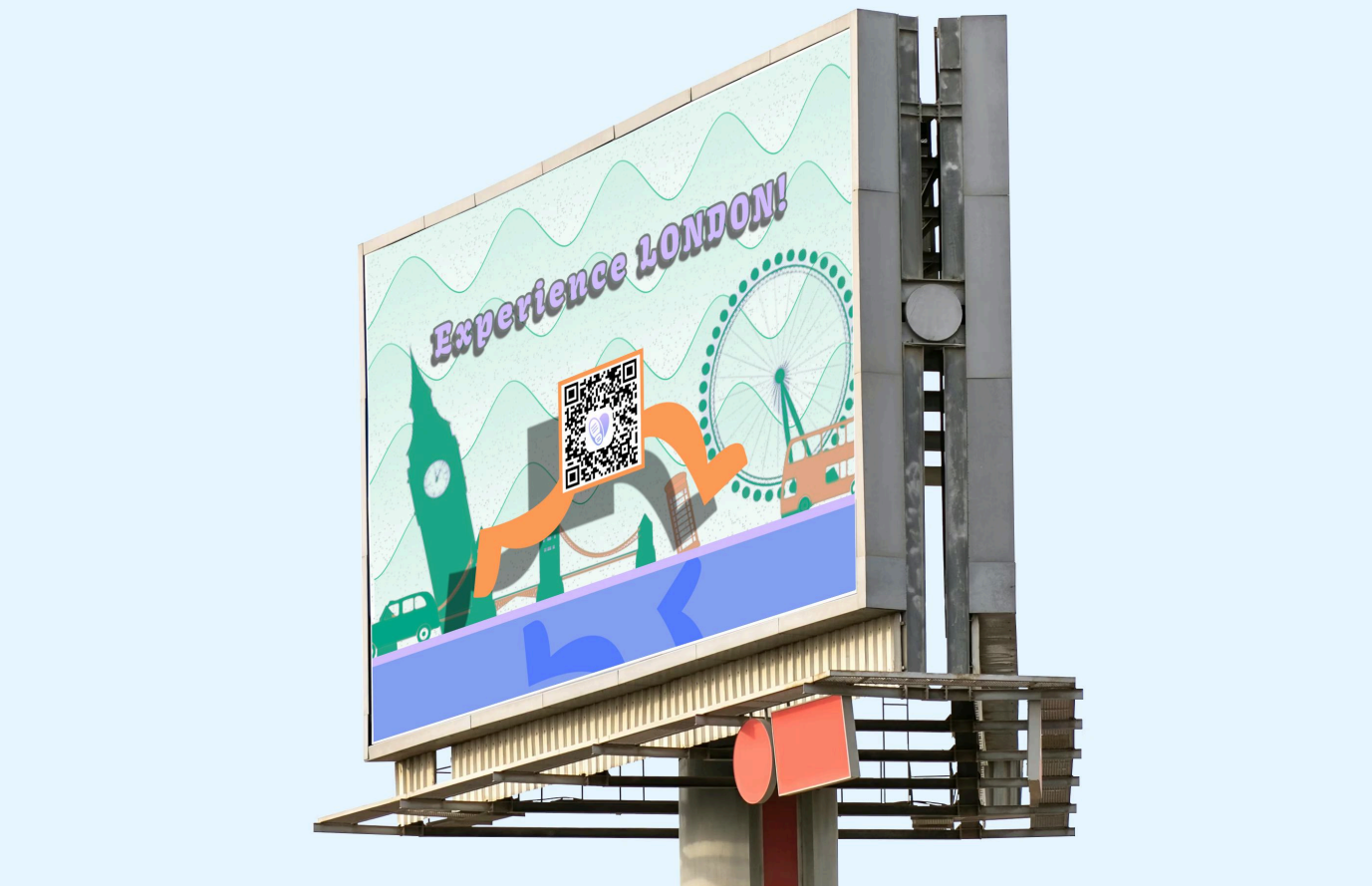


Figure 44 - App search advertising customer acquisition costs

Two Steps potential users are naturally outdoorsy, which is why we expect 30% of downloads to come from Out of Home ads (OOH) [Survey]. Two Steps will primarily be using 48 sheet billboard as it is "the most used outdoor advertising format within the UK" [69]. It costs around £350+vat for a two week display period outside of London.

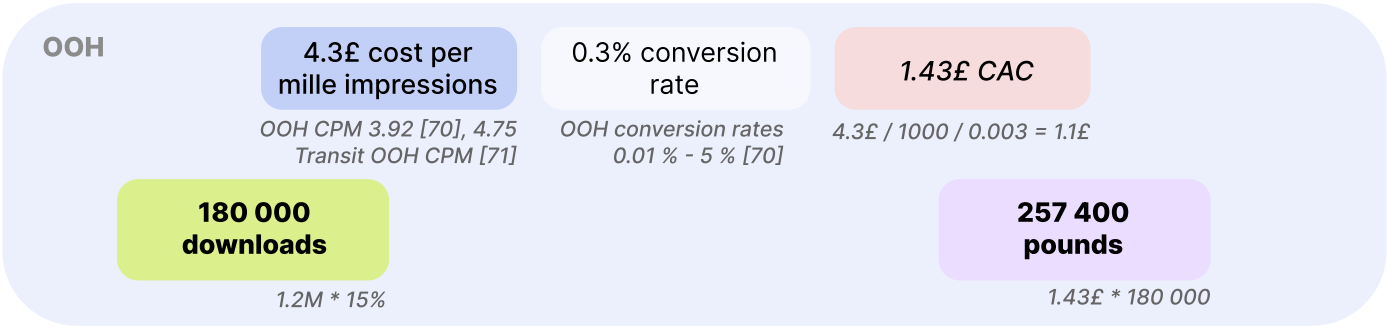


Figure 45 - Total Out-Of-Home advertising costs

9.6 Organic Promotions

The rest of the sales will come organically as the community grows. We consider that this will reach 30% of the promotion of the app.

- This will in part come from user promoting it to each other by sharing their scores and planning together on the app. We also expect devoted users to share their favourite locations through the QR code sharing ability on social media, organically inviting more customers.
- In addition to this, companies will push their customers to follow them on Two Step as they realise that it leads to more repeat customers.
- Tourism companies and other organisations will share go-to locations through Two Step, bringing further customers to the platform.

9.7 Free to Premium account funnel

Within mobile gaming apps 82% of their revenue comes from subscription accounts, with 36% for non-gaming [72]. The general conversion rate to in-app purchases is around 2% on Android and IOS apps [73]. Though it is clear that this varies widely relative to the purchasing advantages and marketing, with Spotify having a 28% conversion rate in Europe [74]. Which is why Two Step markets the premium model within the app using its gamification features and ads.

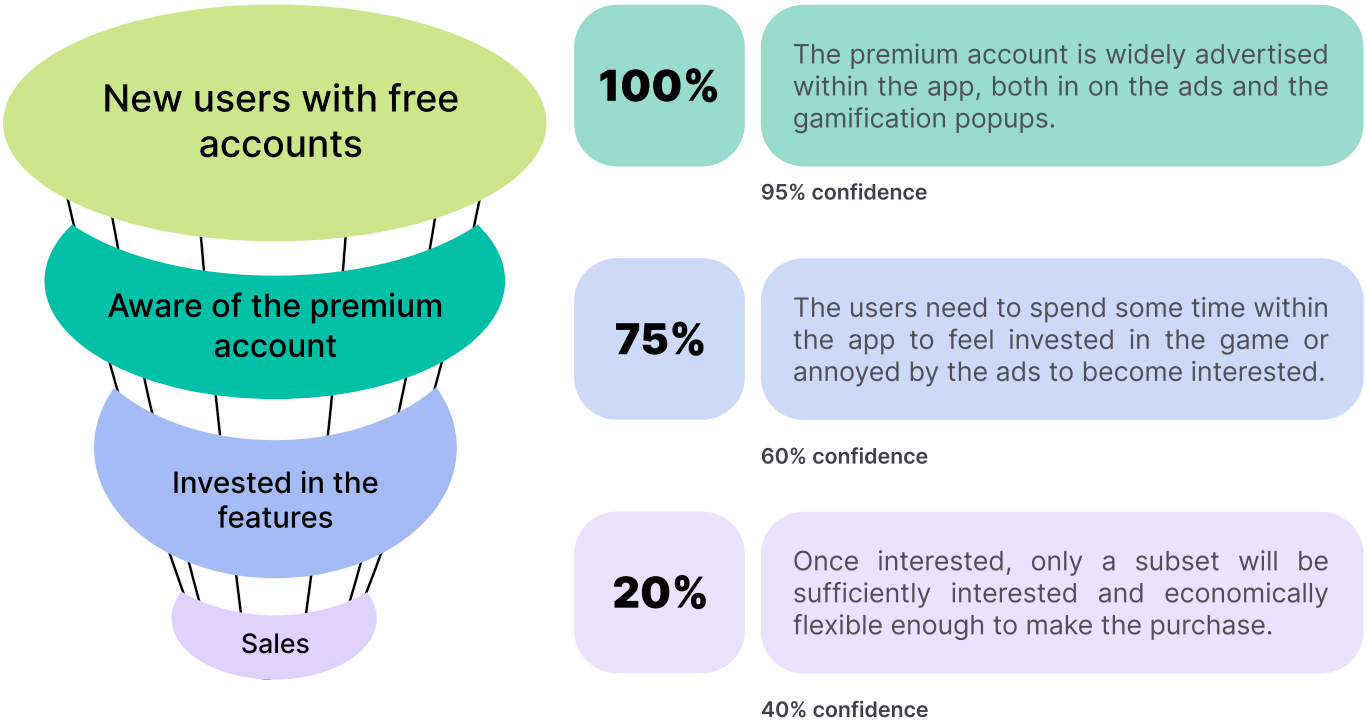


Figure 46 - Free to Premium funnel

The primary bottleneck of the funnel is getting users past the paying threshold. While multiple elements increase the chances of sales (co-creating value [75], virtual goods [76], and social proofs [77]), it is uncertain the qualitative metric of the conversion. From the figure above, we can approximate the customer segments.

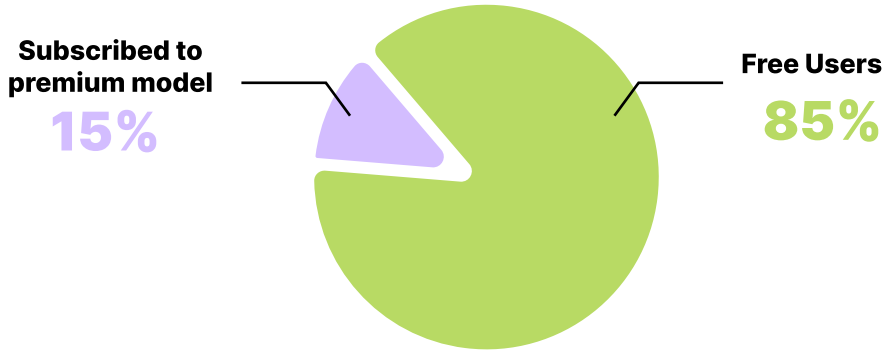


Figure 47 - Free and Premium account pie chart

Considering that Two Step is a non-gaming app while still incorporating gamification elements, we expect a large portion of our revenue to come from premium accounts even though they only represent 15% of our users.

10. Profit & Loss Statements

10.1 Marketing Costs

As a start up, the marketing cost will take a significant portion of our budget. Our innovative approach to leisurely walking wil require a significant exposure and social hype to gain popularity in the extremely competitive walking community.

	Initial Investment	Year 1	Year 2	Year 3	Year 4	Year 5
App Search Ads	£6,247.5	£6,247.5	£6,247.5	£6,247.5	£6,247.5	£6,247.5
Social Media	£9,450	£9,450	£9,450	£9,450	£9,450	£9,450
Outdoors Ads	£4,504.5	£4,504.5	£4,504.5	£4,504.5	£4,504.5	£4,504.5
Organic Word of Mouth	-	-	-	-	-	-
Annual Total	£20,202	£20,202	£20,202	£20,202	£20,202	£20,202

Table 5 - Marketing Costs for First 5 Years

10.1.1 Total Cost of Acquisition

Different sales channel have individual CAC price. To simplify Two Step's financial projections, we are maintaining the CAC consistent for all sales channels during the first year and re-evaluating after to adjust accordingly. Individual CAC and the marketing spend can be used to calculate the customer requisition rates per month for each sales channel.

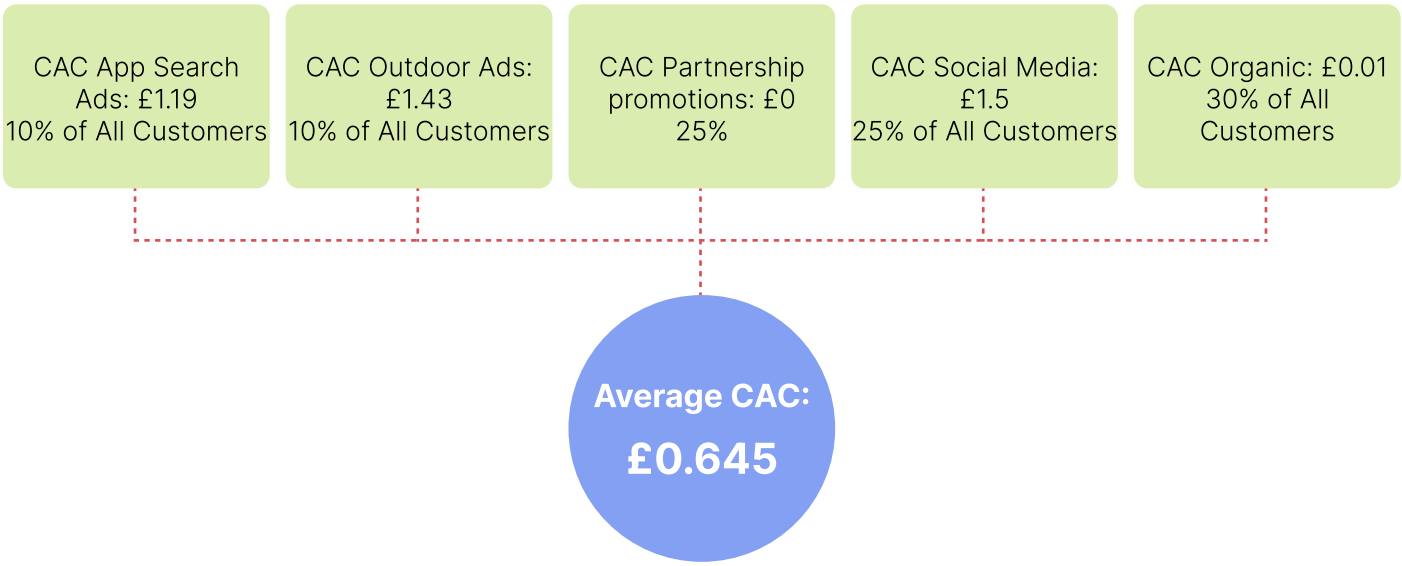
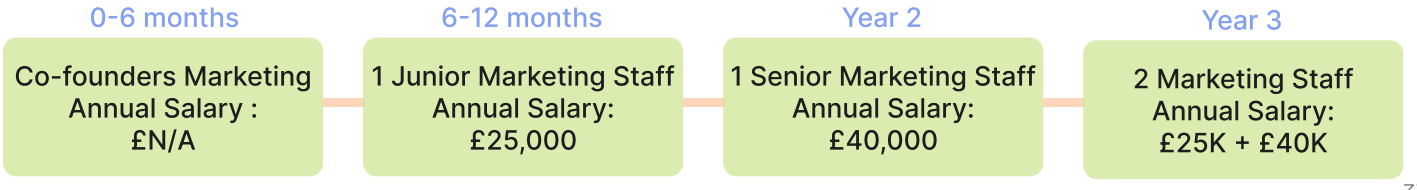


Figure 48 - Two Step's Overall CAC Costs

Our average CAC of £0.645 indicates a healthy, affordable cost of acquiring customers.

10.1.2 Marketing Staff Costs

Two Step's marketing team will initially be comprised of our co-founders during the first 6 months. From month 6, the first junior sales employee will be hired to manage and expand the marketing and sales channel. During the second and third year the sales team will expand to hire a senior and two marketing staff respectively.



10.1.3 Business Partner Acquisition

Although our main users are walkers, we also target individual businesses who want to advertise their services. We want to ensure we can provide walkers with broad activities, local shops and restaurants to explore during their walks. We initially source 30 businesses (around 3 per section in London) to kick-start our user's acquisition journey, then expanding it each year. This process would be cumulative, increasing monthly due to word-of-mouth and outdoor ads. Additionally, our sales channels would act a secondary means of acquisition.

10.1.4 Customer LTV (Lifetime Customer Value)

Our LTV represents the total revenue Two Step expects from a customer throughout their entire engagement, including the different revenue channels: subscription, ad and partnership interactions. Since our LTV exceeding our CAC, it signifies a sustainable and profitable acquisition strategy. During our growth and expansion as a company, we expect our LTV to increase as marketing channels, customer experience and insights in customer behaviour are optimised. To ensure viable growth, rigorous LTV, CAC and marketing monitoring will be conducted.

Subscription User:

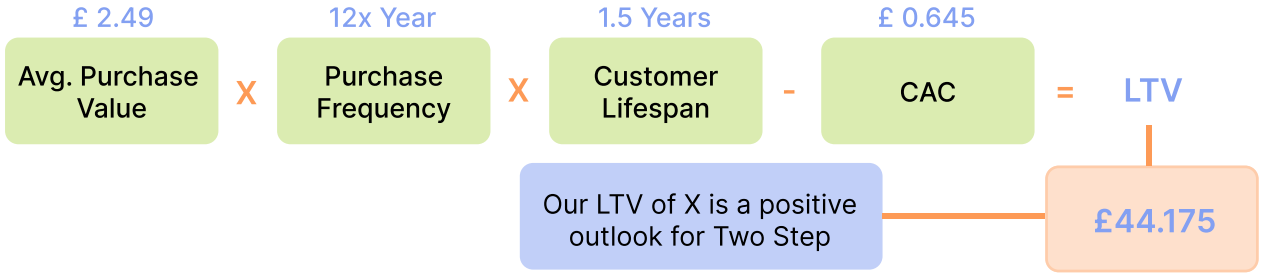


Figure 49 - Subscription Lifetime Customer Value

General Ads:

Figure 50 - General Advertisement Lifetime Customer Value

Partnership Ads:

Figure 51 - Partnerships Advertisement Lifetime Customer Value

Full LTV:

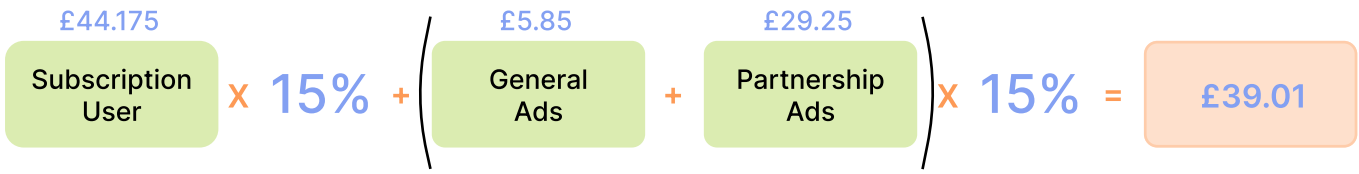


Figure 52 - Full Lifetime Customer Value

10.2 App Development Costs

The Two Step app will consist of four stages: Discovery which usually costs 10-15%, design which costs 20-25%, the development stage would account for the biggest cost around 40-55%. The last stage of testing and deployment would account for 15-20% and 5-10% respectively [78].

10.2.1 Discovery Costs

At this stage, research, market analysis and concept validation are conducted to match the needs of the target audience to app functionality and define business goals. Although this process usually takes between 30-40 hours and usually costs around £5-10k [79], the initial analysis has been validated by this report and will be continued by the CEO, therefore it's accounted for in their salary. However, additional initial user validation should be conducted through further surveys tools, like Qualtrics during 1 month [80].

	Initial Investment
Qualtrics	£450

Table 6 - Qualtrics Survey Tools Costs

10.2.2 Design Costs

The second stage, would focus on the UI/UX design, creating wireframes, prototypes and mock-ups. However the main app outline has already been developed and detailed in this report, so the cost hiring a UI/UX designer can be removed. However, accounting for design software tools is essential as they will be needed to update and maintain the app:

	Initial Investment	Year 1	Year 2	Year 3
Figma or Sketch	£150	£150	£150	£150

Table 7 - Design Software Tools Costs

10.2.3 Development Costs

App developments costs will have the biggest impact on the costs, covering coding, backend infrastructure and integration of features.

Computational Costs

These refer to the costs associated with cloud computing services, server hosting, data storage and processing power. Google Cloud hosting services are used due to reliability and adaptability. Server Infrastructure, Data Storage and Computational Power increase yearly and grow as user base and data increase. Additionally, Google Maps API services would be used for wayfinding. Their costs fluctuate but are assumed to be proportional to the number of users.

Function	Pre-Launch	Year 1	Year 2	Year 3
Cloud Hosting	£2,000	£4,000	£7,000	£10,000
Server Infrastructure	£2,000	£4,000	£6,000	£8,000
Data Storage & Backup	£750	£1,500	£2,500	£3,500
Computational Power	£1,000	£2,500	£4,000	£5,500
Third-party APIs	£1,250	£2,000	£3,000	£4,000
Security	£400	£1,000	£1,200	£1,500
Analytics & Monitoring	£400	£1,000	£1,200	£1,500
Total Cost	£7,800	£17,600	£24,900	£34,000

Table 8 - Computational Costs

General Administration

Legal and accounting administrative tasks during the development phase, such as entity formation, IP protection, licensing, tax compliance, and employee agreements, will primarily be managed by the CEO and CFO. For specialized or one-off requirements, a monthly consultation with an accountant (£250) and solicitor (£200) is assumed, ensuring cost-effective support for compliance and regulatory needs [81][82].

Employees

At the start, employees salaries will account for the greatest costs. Achieving a balance between ethical salaries and budget limitations is essential to ensure that the team is well-compensated while adhering to the Two Step's financial goals. Table X outlines the salaries progression during the set up and following 3 years. Both the Junior Marketing and Software Engineer will have a short term contract as more experienced employees will be integrated when the company grows and experiences an injection of cash in the third year.

Assumptions:

- 5.6% salary increase per year according to average weekly earning in Great Britain [83]
- 12% national insurance
- 3% minimum pension contribution
- Training & onboarding per employee: £500

Employee	Pre-Launch	Year 1	Year 2	Year 3
CEO	£36,000	£38,016	£40,145	£42,393
CTO/Lead Engineer	£36,000	£38,016	£40,145	£42,393
CMO	-	-	£40,145	£42,393
CFO	-	-	£40,145	£42,393
Junior Marketing	-	£25,000	-	£27,878
Software Engineer	£32,000	£33,792	-	£37,682
Customer Service	-	-	£12,500	£13,200
Total Salary Cost*	£119,600	£155,047	£199,042	£248,332

*including pension and NI

Table 9 - Employees Salaries Costs

10.2.4 Testing & Deployment Costs

A testing stage is essential for any app mobile, involving rigorous testing to identify and fix bugs, ensuring the app runs smoothly and securely across multiple devices and conditions. This includes unit, system and regression testing. Costs for this phase depend on the number of tests carried out, usually in the range of £8,000-18,000 [80]. Assuming, 150 unit tests, 50 system tests and 100 regression tests will be carried out during the 2 lasts months of development: [84]

	Number of Tests	Cost Per Test	Cost Per Test
Unit testing	70	£30	£2,100
System testing	25	£75	£1,875
Regression testing	50	£60	£3,000
Total Cost	-	-	£6,975

Table 10 - Testing Costs during 2 months

Lastly, the deployment phase focuses on App Store and Play Store testing, ensures the app compliance with platform requirements (e.g: UI guidelines, data privacy policies)

	Estimated Cost
Deployment	£1,000

Table 11 - Deployment Costs

10.3 Setup Costs

Ensuring Two Step's robustness and performance is a priority. During the 6-month development period, the primary focus will be on discovery, design, development and testing before deploying the app. Within these stages, our CEO will manage initial marketing, design and some minor administrative tasks and our Lead Engineer, supported by one software engineer, will handle the majority of development at the beginning. Since the company will be operating without initial external funding, the budget accommodates only 3 employees at this stage. Following the app's launch, the first three months will concentrate primarily on generating sales and executing marketing strategies.

	6 month Setup
Discovery	£450
Design	£150
Development	£8,250
Employees	£59,800
Testing & Deployment	£7,795
Total Cost	£76,445

Table 12 - Computing Setup Costs

10.4 Future Maintenance Costs

Mobile app development has an additional maintenance component. The industry average accepted cost to maintain software is around 15-25% of its original development cost. The maintaining process cost can be broken down into the following components: [78]

1. **Bug fixing and updating:**
   There will inevitably always be bugs and updating necessary in the app
2. **Security updates:**
   Security breaches still happen and they're impossible to fix permanently
3. **Performance monitoring:**
   Ensuring app performs without glitches to deliver smooth user experience
4. **Third-party API maintenance:**
   Ensuring app is compatible with latest third-party API updates
5. **Platform updates issuing:**
   Adjusting app's code to comply with latest mobile platform update
6. **Server and backend maintenance:**
   Ensuring apps' reliability, performace and scalability
7. **User support and feedback:**
   Ensuring user satisfaction, resolving issues quickly and providing customer support
8. **Design and UI / UX updates:**
   Ensuring app remains modern, intuitive and competitive
9. **Database management:**
   Ensuring data integrity, security and optimal performance
10. **Licenses and subscriptions:**
    Expenses for third-party tools, services and software necessary for the app's operation and enhancement

	Potential Cost
Bug fixing and updates	£1,000-5,000
Security updates	£1,000-2,500
Performance monitoring	£2,400-24,000
Third-Party API maintenance	£3,00-20,000
Platform updates	£5,000-30,000
Server & backend maintenance	£5,000-25,000
User support & feedback	£2,000-10,000
UI/UX updates	£5,000-25,000
Database management	£5,000-15,000
Licenses & subscriptions	£1,000-10,000

Table 13 - Computing Setup Costs per Year

Although, the maintenance costs don't influence the initial 6 month setup costs, they provide insights into what the future expenses and necessary revenue for the app might be. Considering our app is a medium level sophistication app, we can assume the potential cost to be in the middle rang, and so the future maintenance costs will be around: £98,450.

10.5 Cost of Goods Sold (COGS)

For the Two Step App, unlike traditional products, there are no physical goods, but there are still ongoing costs associated with digital products with delivering systems. Therefore, the COGS would translate to operational and maintenance costs that keep the app functioning. This was further analysed during the future maintenance costs. (10.4)

10.6 Sales & Scenario Analysis

The sales and proposed scenarios for worst, best and expected outcomes for Two Step can be estimated by taking assumptions to support scenarios. Stress testing was explored with sales assumptions to uncover Two Step's vulnerabilities.

Subscriptions

Based on marketing expenses and customer acquisition, number of subscribed user can be estimated and revenue generated can be calculated.

Ads Revenue

Based on advertisement pricing and user assumptions the revenue can be calculated.

Business Partnerships

Based on assumptions made on number of partnership with business made and

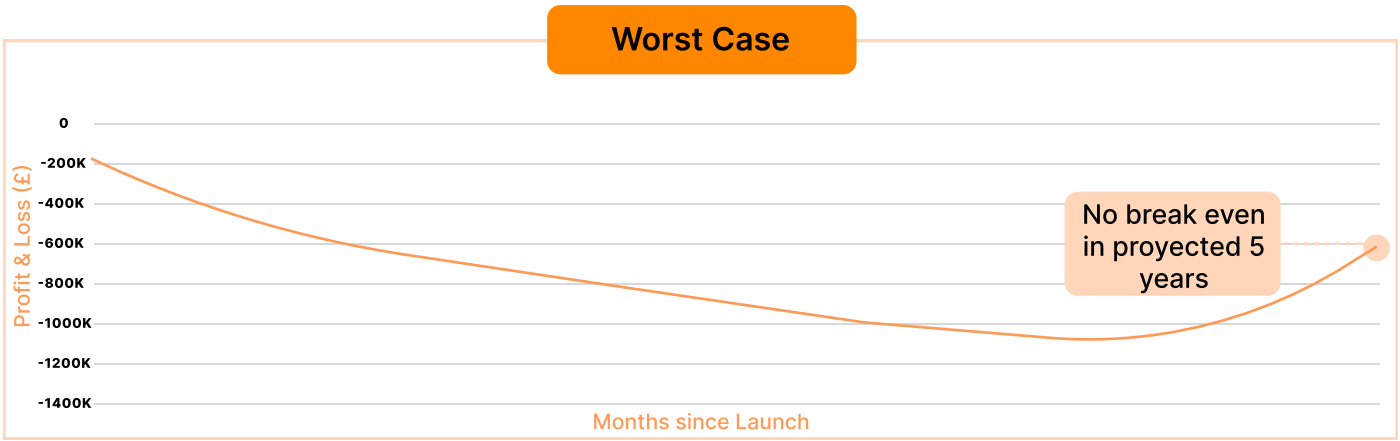
Stress Testing

Three potential scenarios are analysed in Table 14 and forecasted P&L statements were made for each case. Additionally, graphs in Figure 53 show the cumulative P&L and their respective break even points, which give us an insight on Two Step's minimum need for survival.

	Worst	Expected	Best
Subscribed users	150k	180k	210k
Subscription Price	£1.5	£2.49	£3.5
Free users	850k	1.02M	1.19M
Ads Price*	£0.1	£0.3	£0.5
Partnerships Ads Price*	£0.3	£0.5	£0.7
Annual Revenue	£3.20M	£6.86M	£11.5M
Confidence Factor	85%	70%	60%

Table 14 - Stress Testing with Key Assumptions

*price per click



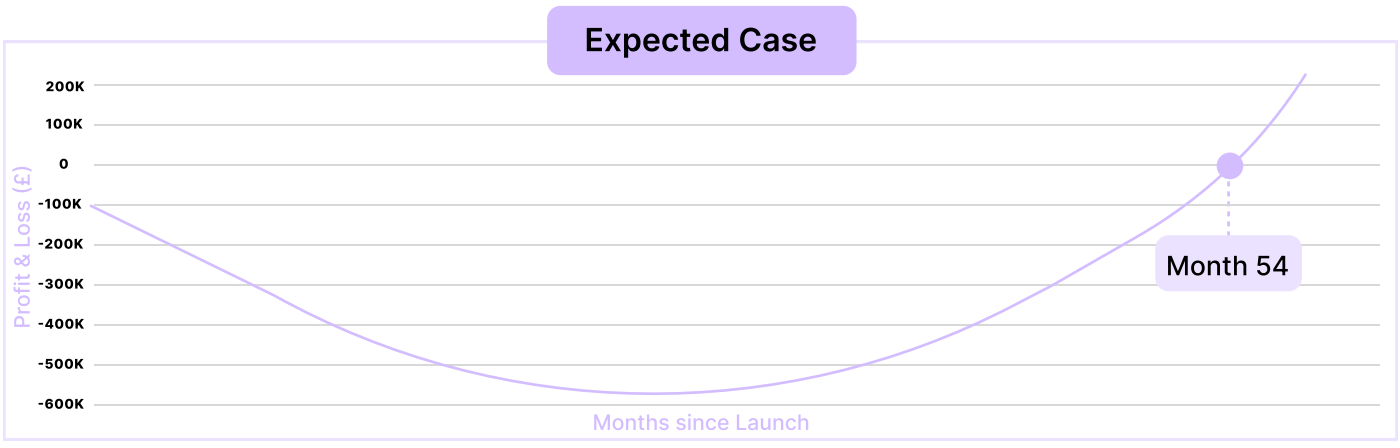


Figure 53 - Two Step's Projected Cumulative P&L for Initial 5 Years

See Appendix E for full calculations

10.7 Net Profit or Loss

Given the high costs associated with setting up and gaining a large enough customer base for viability, the projection looks weaker than ideal with a projected break even point at 54 months. An initial investment of £582k is estimated to be sufficient to support Two Step during the 6 month development and user acquisition phase. Once the high starting costs have been covered there is however a steep projected rise into profit. Nearing the end of a 5 year projection Two Step is estimated to be worth just over £250k

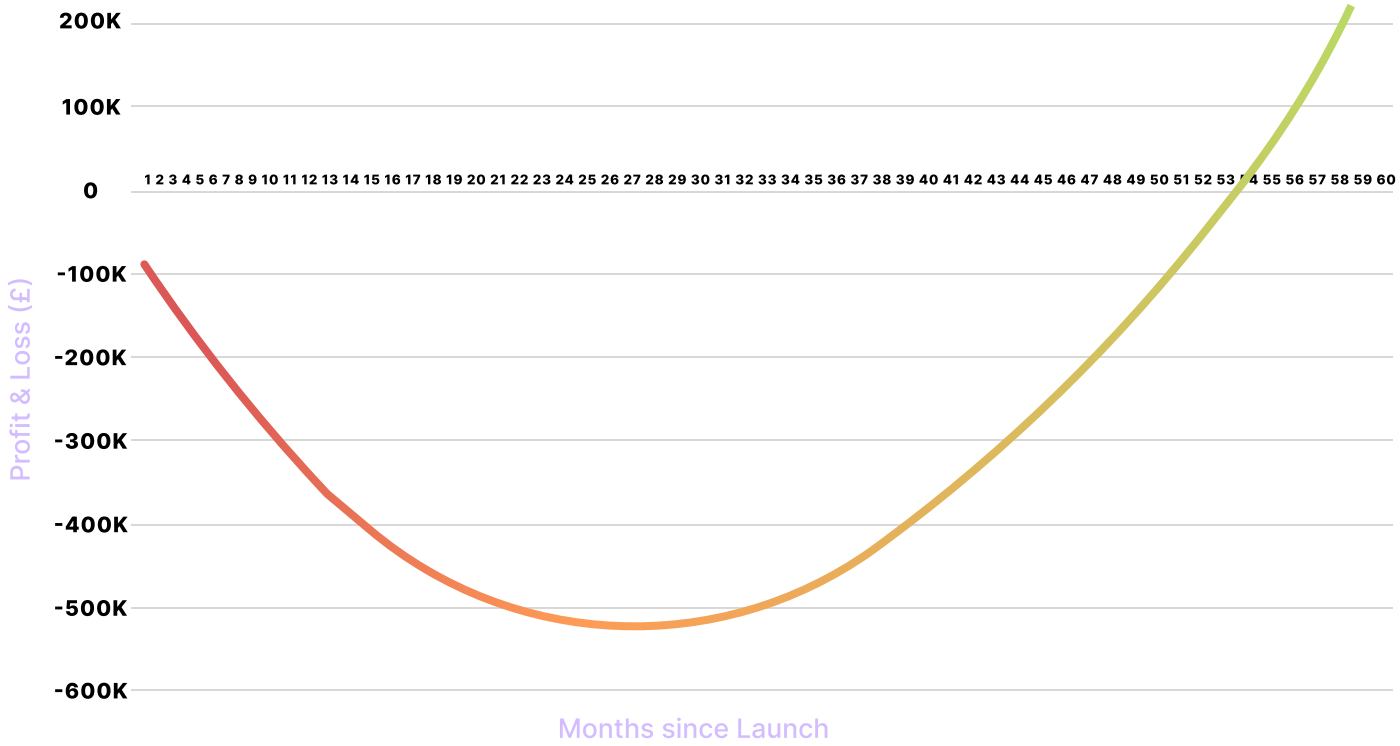


Figure 54 - Two Step's Projected Accumulative Profit & Loss in the initial 3 Years

10.8 Summary

A high level summary of the anticipates profits and losses are summarised below. A good indicator of a successful company is an EBITDA margin of around 30% [85]. By our fourth year in business we wil have an **EBITDA margin of 34%** which will place as a growing and stable company. As Two Step grows in size it will look to expand to further cities, requiring additional employment and resources, however overall this will further boost revenue.

	Pre-Launch	Year 1	Year 2	Year 3	Year 4
Revenue	£0	£258,300	£362,880	£497,340	£580,100
Cost of Goods	£0	-£98,459	-£108,304	-£119,135	-£131,048
Gross Profit	£0	£159,841	£254,576	£378,205	£449,052
Gross Margin	-	62%	70%	76%	78%
Sales & Marketing	£20,202	£20,202	£20,202	£20,202	£20,202
Discovery	£450	-	-	-	-
Design	£150	£150	£150	£150	£150
Development	£8,250				
Testing	£6,795	£6,795	£6,795	£6,795	£6,795
Deployment	£1,000	£1,000	£1,000	£1,000	£1,000
Employees	£59,800	£155,047	£199,042	£248,332	£262,239
EBITDA	-£56,243	-£40,953	£2487	£67,729	£153,966
EBITDA Margin	-	-26%	1%	18%	34%
Corporate Tax	£0	£0	£63,644	£94,551	£112,263
Net Income	-£56,243	-£40,953	-£61,157	-£26,822	£41,000

Table 15 - Two Step's Projected P&L Statement

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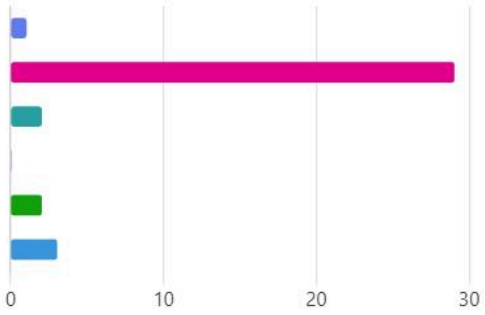
APPENDIX

Appendix A: Survey Questions & Results

1. How old are you?

More details

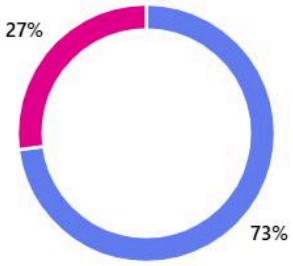
Under 18	1
18-24	29
25-34	2
35-44	0
45-54	2
55+	3



2. What gender do your identify with?

More details

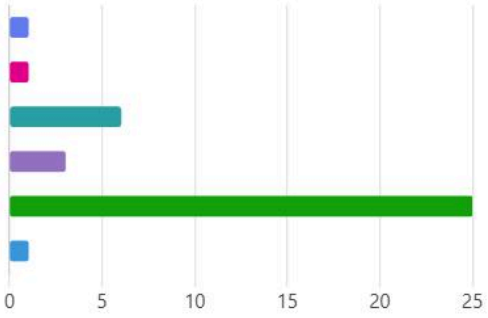
Woman	27
Man	10
Non-binary	0
Prefer not to say	0



3. What type of area do you live in?

More details

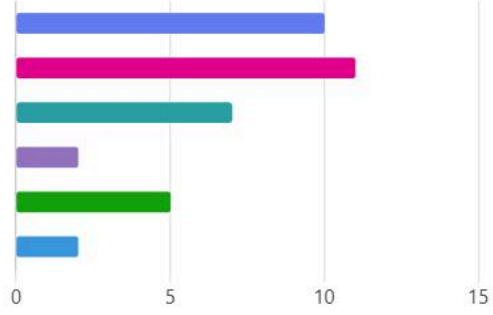
Countryside	1
Village	1
Town	6
Suburbs	3
City Centre	25
Other	1



4. How often do you spend time walking leisurely?

More details

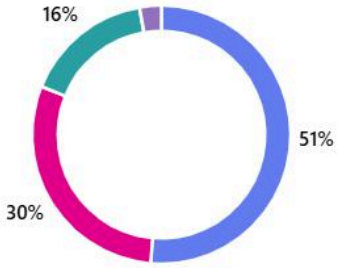
Daily	10
2-3 times a week	11
Once a week	7
Once a month	2
Rarely	5
Never	2



5. How organised are your walks?

More details

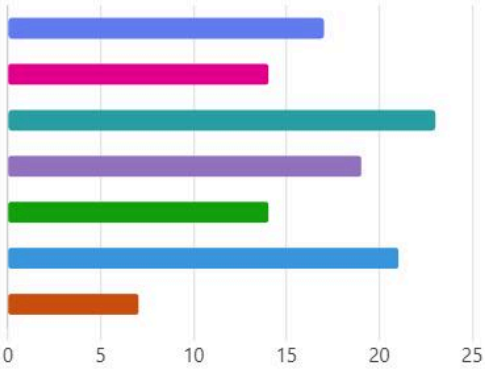
Fully spontaneous	19
Quick search	11
Completely planned	6
Other	1



6. Where do you usually walk to? (Select all that apply)

More details

Cafe	17
Restaurant	14
Park	23
Shops	19
Work	14
Just walking around	21
Other	7



7. When you are going out how often do you like trying new places?

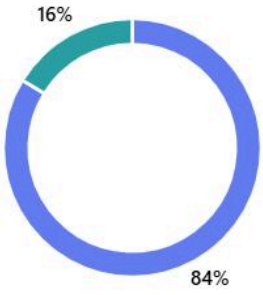
More details

Promoters	8
Passives	15
Detractors	14

8. Would you like to try new places?

More details

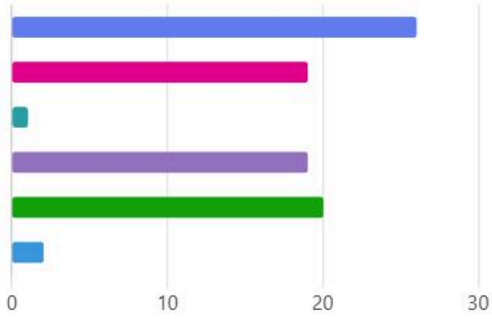
Yes	31
No	0
Maybe	6



9. When trying new places, how do you find them? (Select all that apply)

More details

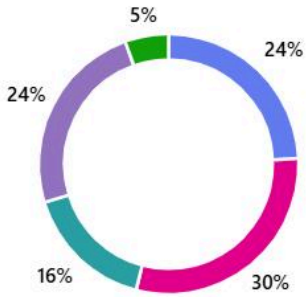
Social media	26
Internet search	19
Blogs	1
Word of mouth	19
Navigation apps i.e. Google Maps	20
Other	2



10. Do you enjoy planning your walks?

More details

Yes, I enjoy it	9
Yes, but I find it tedious/time-consuming	11
No, I don't enjoy it	6
I don't plan my walks	9
I follow someone else's route	2



11. How satisfied are you with current navigation apps?

More details

Promoters	5
Passives	21
Detractors	11

12. What features of current navigation apps do you not find satisfactory?

More details

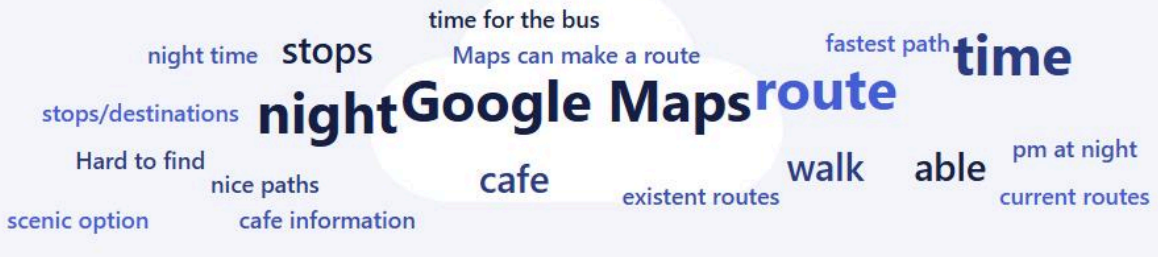
15
Responses

Latest Responses

"I wish ar mode was more reliable and worked more often. Also I don't like w... "

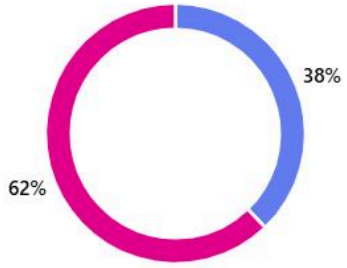
• • •

3 respondents (20%) answered Google Maps for this question.



13. Have you ever looked for/downloaded apps to help you on your walks? (e.g. All Trails) More details

Yes 14
No 23



14. We're designing an app for leisurely walks, what features would you enjoy? (Eg: the ability to personalise your walking route) More details

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Responses

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"I don't want one. Maybe you should have an app that rewards exploration, li... "
"Tracking time and locations visited"
"personalise routes and clearer roads"
...

10 respondents (38%) answered routes for this question.

15. If you are interested in this project, and would be open to answering more questions, please leave your email below. More details

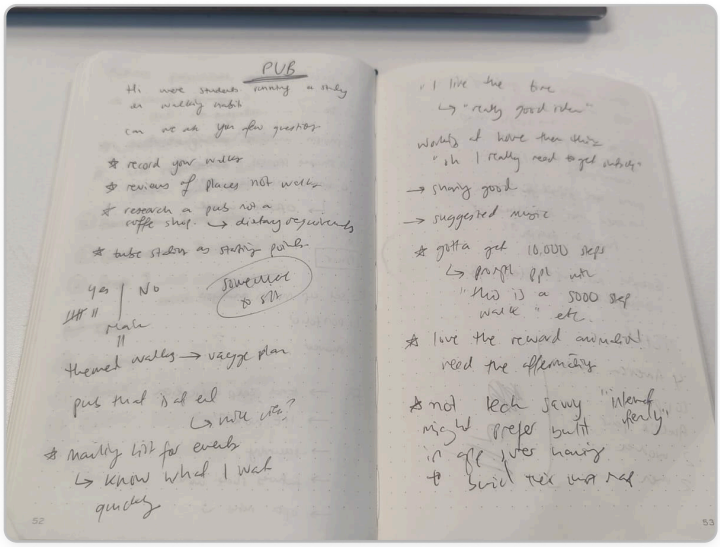
15
Responses

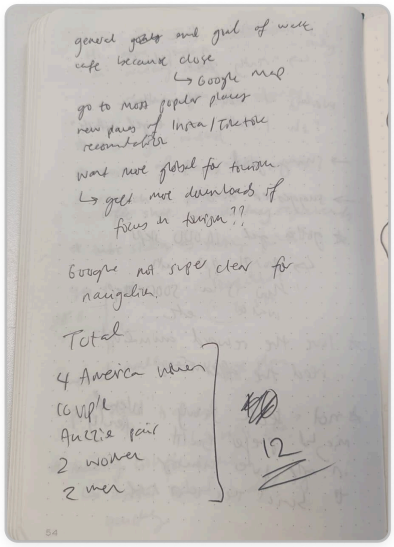
Latest Responses
"theharveygould@gmail.com"
"wellokayemmalin@gmail.com"
...

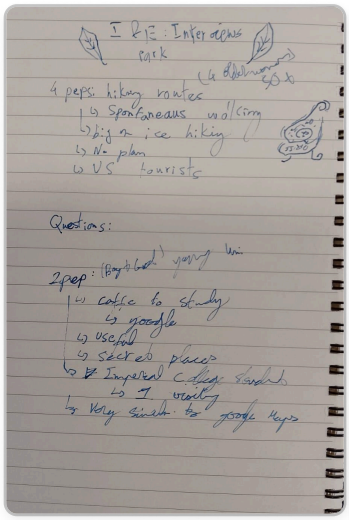
1 respondents (7%) answered species in parks for this question.

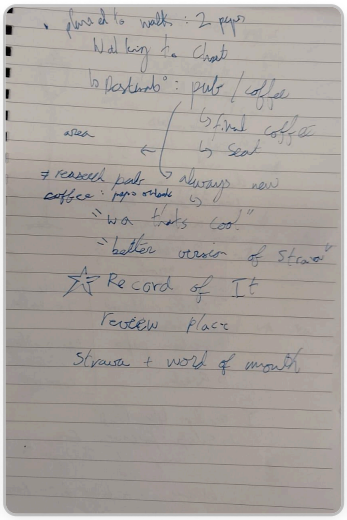
Appendix B: Park Interviews Questions & Insights

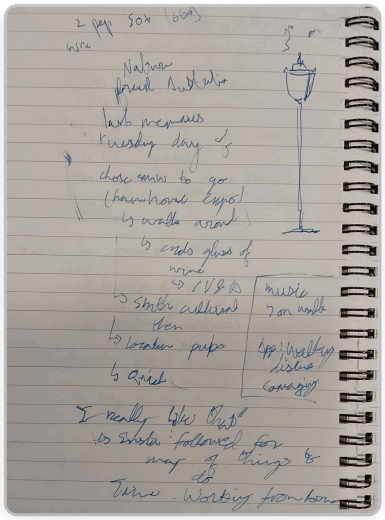
Question 1	What is your motivation for your walk?
Question 2	Did you plan your walk today?
Question 3	Have you visited any places on your walk today?
Question 4 (showing prototype)	Do you think you'd benefit from an app like this? What would you change? What other features would you like on it?
Question 5	Would/do you or do you not try new places?

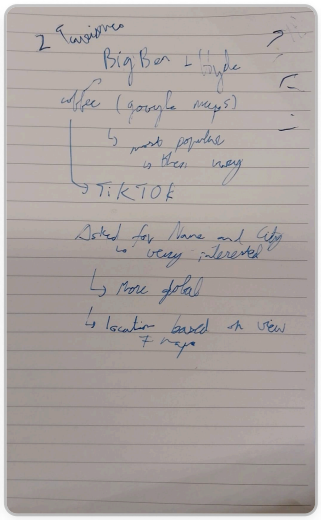












Appendix C: Assumption ranking

External Internal

Assumption description	Uncertainty score (out of 10)	Justification	Impact score (out of 10)	Justification	Total risk score (uncertainty × impact)
Phones and apps will continue to be widely used	1	Smartphone adoption continues to grow globally, ensuring app usage remains relevant	6	If phones are no longer used, we would have to pivot to another media	6
In the coming years, there won't be another major pandemic, or another disruption, that prevents people from walking for leisure	3	Pandemics are rare, but climate-related events or geopolitical issues aren't inconceivable either	4	While it could temporarily affect user activity, the app would remain relevant in the long term	12
The app will be readily available and easily downloadable on all major app stores	2	App distribution via app stores like Google Play and Apple Store is relatively straightforward, but approval processes could delay launch	7	Delays or inability to launch on app stores would severely affect user acquisition and trust	14
Users of all types will want features like curated itineraries, and recommended places	4	Preferences vary widely, some users might find these features unnecessary	5	Broad appeal is critical for scaling. Even if most users were to find these features unnecessary, Two Step still offers other functionalities which appeal to everyone	20
Walkers don't always have a specific destination in mind (a specific restaurant, coffee shop, pub, etc.)	4	People often have broad preferences but, according to our survey, are open to trying new things	6	If users always have a specific destination in mind, the app's core functionality of randomised exploration loses appeal. However, the app could still pivot	24
Users value customisable routes that reflect their preferences and avoid repetitive paths	3	Personalisation is highly valued in apps	8	Customisation is one of the app's key differentiators. In this case, the app would lose a significant competitive advantage	24
Walkers want to be more spontaneous with their walks	5	While some enjoy spontaneity, our user research showed others prefer fixed plans and destinations	5	If most users prefer fixed routes, the app could still cater to them as every route can be saved as a QR code	25
Beachhead customers will be willing to pay a small subscription fee for premium features	3	Many navigation apps already rely on subscriptions for monetisation	9	If customers don't pay, it could drastically limit revenue potential. However, it is not our only revenue stream	27
The app's UI will be intuitive enough for seamless onboarding	4	Intuitive design is achievable, though it may require additional user testing	7	Poor UI design would deter users and impact retention	28
Small local businesses in London continue to exist, standing their ground against corporate chains	5	The trend toward corporate chains is strong, but London does have a unique focus on local culture and small businesses	6	If local businesses decline, the app's focus on promoting them becomes irrelevant, but the app could still shift to partnering with larger businesses or pop-ups and events	30
The app's gamification features will increase user engagement and retention	5	Gamification has proven to be effective in enhancing user engagement in other apps (Duolingo), further validation is required for our specific app	6	If gamification fails, user engagement could drop, but the core app can still succeed	30
There is a large enough audience that prefers exploration over efficiency	4	While exploration is a niche, apps like Pokémon Go and Komoot have proven that there is interest	8	This would greatly impact the app's core value proposition, the app's niche depends on attracting exploration-minded users who want something different from Google Maps or Citymapper	32
Users will find real-time nearby activity recommendations valuable and reliable	5	Real-time recommendations are valuable but depend heavily on accurate data, which is reliant on external sources	7	If recommendations are inaccurate, it damages user trust and therefore retention	35
Generating randomised yet customised routes is technically feasible	7	Proven algorithms and technologies exist for efficiency focused routing (optimising distance) but there is little evidence of elements of randomness being added, or aiming for a longer route	6	If technical challenges arise, they could significantly delay the app's launch or require modifications to the concept	42
Mapping data (e.g. location of open businesses) is accurate and readily available	5	Most navigation apps we have researched use their own collected data, we have yet to find a source for our app	9	The app relies on mapping data for functionality. Inaccurate or unavailable data would render key features unreliable.	45
Local businesses will pay for exposure or partnerships to attract app users	4	Prior research has shown that local businesses often pay for ads, which suggests that they would be willing to invest in our app, though this is subject to costs and app popularity	9	If businesses are unwilling to pay, a major revenue stream vanishes	36
Users trust the app's ability to protect their data and maintain privacy	6	Privacy concerns are a growing issues amongst users, especially with location-based apps	8	Privacy concerns could deter users, leading to low adoption rates and regulatory scrutiny	48

Appendix D: Sales Analysis

Overall revenue profits were calculated using the following assumptions:

- subscribed users = 15% of 1.2M (total users) = 180K
- free users = 85% of 1.2M = 1.02M
- walks taken weekly = 5 (from survey)

- Subscribed revenue = Subscribed users x Subscription Price

- Total Revenue Ads 1 = Free users x N times walked weekly x 4 x N weeks per month (4) x 5% [X] x Partnership Price

- Total revenue ads 2 = Free users x N times walked weekly x 4 weeks per month x 1 ads x 5% [X] x Partnership price

- Total revenue = subscribed rev + total rv ads 1 + total rev ads 2

- total expenses = 6 month set up costs+ salaries year 1+ maintenance

- Final profit per year = total revenue-total expenses

	A	B	C	D
1		Worst	Expected	Best
2	subscribed users	150000	180000	210000
3	Subscription price	1.5	2.49	3.5
4	Subscription rev	2700000	5378400	8820000
5				
6				
7	non subscribed users	850000	1020000	1190000
8	ads Price	0.1	0.3	0.5
9				
10	Total rev ads 1	85000	306000	595000
11				
12	partner price	0.3	0.5	0.7
13	Total rev ads 2	765000	1530000	2499000
14				
15	total rev	3550000	7214400	11914000
16				
17	minus expenses			
18	setup costs	76445	76445	76445
19	salaries for 1 year	155047	155047	155047
20	marketing	20202	20202	20202
21	maintenance	98450	98450	98450
22	total expenses	350144	350144	350144
23	new total	3199856	6864256	11563856

[X] Average Click-through rate. (Apr 25, 2024) Available at: https://thesocialshepherd.com/blog/google-ads-ppc-statistics#:~:text=The%20Average%20Click%2DThrough%20Rate,Google%20Ads%20is%204%2D6%25&text=Data%20shows%20the%20average%20CTR,ads%20is%204%2D6%25.

Appendix E: Expected Case Scenario Analysis

Assuming constant revenue:

- Revenue per user = $\frac{\text{annual revenue}}{\text{total number users}} = \frac{£7214400}{1200000} = £6$
- Initial 6-month setup cost: £76k
- Yearly expenses: £253k
- Expected expenses in 2 years: 76k+253k+253k = £582K
- Necessary users achieved by 2 years: $\frac{\text{yearly cost}}{\text{revenue per user}} = \frac{£253k}{£6} = 42K$
- Users needed per year: $\frac{42K}{2 \text{ years}} = 21K \text{ users/per year}$